



Department of Business and Management

Bachelor's Degree in Management and Computer Science

**ANALYSIS OF STOCK CORRELATION
NETWORKS THROUGH GRAPH TRAVERSAL
ALGORITHMS**

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Abstract

This thesis investigates whether classical graph traversal algorithms and graph-based filtering techniques can reveal interpretable structural patterns in stock correlation networks. The primary empirical analysis uses 47 S&P 500 stocks across ten GICS sectors and daily adjusted closing prices over 2019–2025, yielding 1,759 aligned daily logarithmic-return observations. Pearson correlation networks are constructed across $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$, with $\tau = 0.5$ as the main threshold, alongside a threshold-free Minimum Spanning Tree (MST). Breadth-First Search, Depth-First Search, connected-component detection, and Tarjan-based articulation-point and bridge routines are used to characterise network structure.

Mean within-sector correlation (+0.575) is substantially higher than cross-sector correlation (+0.358). The threshold network fragments non-linearly: the giant component contains more than 95% of the 47-stock universe through $\tau = 0.5$ but falls below one third at $\tau = 0.6$. At the main threshold, BFS from AAPL reaches 44 of the other 46 stocks within four hops, while three articulation points and four bridges identify unique structural dependencies. The MST also exhibits pronounced sectoral organisation, with 56.5% of its edges joining same-sector stocks compared with an 8.2% random-pair benchmark. An illustrative network-diversified portfolio has lower in-sample internal correlation than a sector-diversified benchmark, but this does not constitute evidence of a profitable investment strategy.

A supplementary extension scales the analysis to 474 stocks across 11 GICS sectors. Similar clustering and same-sector edge proportions remain visible at the main threshold, while stricter thresholds produce increasingly sector-concentrated surviving links. The extension also identifies selected stocks whose correlation neighbourhood differs from their formal sector classification.

Overall, classical graph algorithms provide a transparent, reproducible and interpretable framework for examining estimated equity correlation structure. The results describe statistical co-movement rather than causal relationships or predictive investment signals.

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Chapter 1

Introduction

The increasing availability of financial data and the growing influence of network science have opened new avenues for studying the structure of equity markets. Financial markets are inherently relational systems: the behaviour of individual stocks is not independent, but is shaped by a web of correlations and interdependencies that connect assets across sectors and geographies. Representing these relationships as a network—where nodes are stocks and edges reflect the strength of pairwise return correlations—makes it possible to apply tools from graph theory to study market structure in a principled way. This thesis focuses on one specific aspect of that approach: the application of graph traversal algorithms, namely Breadth-First Search (BFS), Depth-First Search (DFS), and connected component detection, to stock correlation networks constructed from historical equity return data. The aim is to examine what these algorithms can reveal about the connectivity and organisation of equity markets, and to discuss the interpretive value of a graph-algorithmic approach for financial network analysis.

1.1 Background and Motivation

Financial markets are among the most complex systems studied in the social sciences. They involve a large number of participants whose decisions interact continuously, producing aggregate outcomes that no single actor controls or fully anticipates. For much of the twentieth century, academic finance represented assets primarily through statistical quantities such as expected returns, variances and covariances. These models explicitly account for dependence between assets, but they do not represent the market as an explicit relational structure of nodes and links. The portfolio selection framework of Markowitz (1952) is the canonical example: it provides a rigorous approach to diversification through the covariance structure of asset returns, without expressing that structure as a graph.

This purely statistical view has well-known limitations. During the 2008–2009 financial crisis, losses spread rapidly across institutions and asset classes through channels that standard risk models had not anticipated. The crisis demonstrated that interconnectedness—

who is linked to whom, and how tightly—matters at least as much as the individual risk profile of each actor (Battiston et al., 2012). In the years that followed, researchers began applying network analysis to financial data more systematically, representing markets as graphs in which nodes are assets or institutions and edges encode the relationships between them.

Within this broader agenda, stock correlation networks have attracted particular attention. In a stock correlation network, each node represents an individual equity, and edges are drawn between pairs of stocks whose return time series are sufficiently correlated. Studies by Mantegna (1999) and Onnela et al. (2003) showed that such networks are not random: they exhibit clear structural patterns, including clusters of stocks from the same economic sector and a hierarchical organisation that reflects the taxonomy of the broader market. These findings suggest that network filtering can make economically meaningful structure embedded in the correlation matrix easier to identify and interpret, particularly sectoral and hierarchical patterns.

This thesis is situated at the intersection of these two areas. Its aim is to construct stock correlation networks from historical equity return data and to analyse their structural properties using graph traversal algorithms—specifically Breadth-First Search (BFS), Depth-First Search (DFS), and connected component detection. The practical implementation is carried out in Python using the NetworkX library (Hagberg et al., 2008). The goal is not to develop a new financial model, but to demonstrate how fundamental algorithmic tools from computer science can be applied systematically to explore the structure of financial networks, and to discuss what the resulting structural findings reveal about the organisation of equity markets.

1.2 Stock Correlation Networks: Key Concepts

A stock correlation network is typically built from a time series of daily equity returns. Given a set of N stocks, the Pearson correlation coefficient is computed for every pair, producing an $N \times N$ correlation matrix C whose entries C_{ij} measure the co-movement between the return series of assets i and j (Mantegna, 1999). Since the full matrix contains $O(N^2)$ pairwise values, a filtering step is usually applied to reduce noise and retain only the most meaningful connections.

The most widely used filtering method is the Minimum Spanning Tree (MST), introduced in this context by Mantegna (1999). The MST converts correlations into distances using the transformation

$$d_{ij} = \sqrt{2(1 - C_{ij})}$$

and extracts the spanning tree of minimum total weight. The result is a sparse graph that captures the dominant correlation backbone of the market. An alternative approach—also

used in this thesis—applies a simple threshold: an edge between stocks i and j is retained only if

$$|C_{ij}| \geq \tau.$$

This threshold-based method is more flexible and allows the analyst to study how the network structure changes as τ varies (Onnela et al., 2003).

Empirical research has established several consistent findings about stock correlation networks. They tend to exhibit a non-random topology: stocks in the same sector or industry cluster together, and the hierarchical structure of the network often mirrors accepted economic classifications (Bonanno et al., 2004). The correlation structure also changes over time, with networks becoming denser and more tightly connected during market downturns (Onnela et al., 2003). These properties make stock correlation networks a useful object of study for understanding how the equity market is organised and how it responds to changing conditions.

1.3 Graph Theory as a Framework for Network Analysis

Graph theory provides the mathematical language for representing and analysing relational structures. A graph $G = (V, E)$ consists of a set of vertices V and a set of edges E connecting pairs of vertices. When edges carry numerical weights—as in a stock correlation network where weights reflect correlation strength—the structure is called a weighted graph. The fundamental vocabulary of graph theory, including concepts such as degree, path, connected component, and spanning tree, enables a precise and systematic description of network topology (Diestel, 2017; West, 2001).

The broader relevance of graph theory lies in the fact that network representations are useful across many domains. Social networks, biological regulatory networks, the internet, and financial systems all share the property that interactions among components shape collective behaviour in ways that cannot be read off from the components individually (Newman, 2010; Barabási and Albert, 1999). In financial markets specifically, graph-theoretic tools have been applied to identify sector groupings, measure market integration, and assess the risk of contagion (Mantegna, 1999; Battiston et al., 2012).

Several structural properties are particularly relevant for the analysis conducted in this thesis. The degree of a node—the number of edges incident to it—indicates how many other stocks a given equity is significantly correlated with. The connected components of a graph identify groups of stocks that are mutually reachable through the correlation network, while isolated nodes correspond to stocks with no significant correlation to any other. Path lengths between nodes, computed by traversal algorithms, capture the topological distance between stocks and help characterise how directly or indirectly different parts of

the market are linked. These are the properties that BFS, DFS, and connected component algorithms are designed to reveal, and they form the analytical basis of the empirical analysis in Chapter 4.

1.4 Graph Traversal Algorithms: BFS, DFS, and Connected Components

Graph traversal algorithms are among the most fundamental tools in computer science. Their function is to visit all the vertices and edges of a graph in a systematic order, enabling the extraction of structural information that is not directly visible from the graph’s adjacency representation (Cormen et al., 2009; Sedgewick and Wayne, 2011). This thesis focuses on three such algorithms: BFS, DFS, and connected component detection.

Breadth-First Search (BFS) starts from a chosen source vertex and visits all adjacent vertices before moving to vertices that are two hops away, then three hops, and so on. This layer-by-layer exploration produces the shortest-path distances from the source to all reachable nodes. Applied to a stock correlation network, BFS makes it possible to measure how many correlation-link steps separate any two equities, and to compute summary statistics such as the average path length and the diameter of the network. Short average path lengths are characteristic of well-integrated markets, where most stocks are reachable from any starting point within a small number of steps (Watts and Strogatz, 1998).

Depth-First Search (DFS) takes the opposite approach: it follows a single path as far as possible before backtracking and exploring alternatives. In the process, DFS produces a spanning tree of the graph and classifies edges into categories that reveal the cycle structure of the network. One practically useful output of DFS is the identification of articulation points—vertices whose removal would disconnect part of the graph. In a stock correlation network, an articulation point corresponds to an equity that acts as a structural bridge between otherwise disconnected clusters. Identifying such nodes has a useful structural interpretation: they represent stocks that occupy an intermediary position in the estimated correlation network, linking parts of the graph that would otherwise become disconnected.

Connected component analysis determines which parts of the graph are internally connected. In an undirected graph, a connected component is a maximal subset of vertices such that every pair is linked by at least one path. When a threshold-based filtering method is used, the component structure of the resulting network depends on the chosen threshold τ : a low threshold produces a densely connected graph with few components, while a high threshold yields many small, isolated clusters. Tracking how components form and dissolve as τ varies gives a practical picture of how cohesive or fragmented the correlation structure of the equity market is.

All three algorithms run in $O(|V| + |E|)$ time and are directly available in the NetworkX

library, making them well suited for exploratory structural analysis of financial networks at the scale of a typical stock index (Hagberg et al., 2008; Cormen et al., 2009).

1.5 Research Gap

The existing literature on financial networks is extensive and methodologically rich. A significant body of work focuses on measuring the topological properties of stock correlation networks—including degree distributions, clustering coefficients, and hierarchical structure—using tools such as the MST and the Planar Maximally Filtered Graph (Mantegna, 1999; Tumminello et al., 2005; Onnela et al., 2003). A separate, more recent strand of research applies network analysis to systemic risk, studying how financial distress propagates through interbank and exposure networks (Gai and Kapadia, 2010; Billio et al., 2012). A third direction uses community detection algorithms to identify sector groupings and to inform portfolio diversification strategies (Fortunato, 2010; Peralta and Zareei, 2016).

What is less common in the literature is the use of graph traversal algorithms—BFS, DFS, and connected component detection—as primary analytical tools for exploring the structure of equity correlation networks. These algorithms are well known in computer science and are introduced in virtually every textbook on the subject (Cormen et al., 2009; Sedgewick and Wayne, 2011). Recent financial-network research continues to emphasise aggregate topological and network-based indicators—for example, threshold networks, Minimum Spanning Trees and network-centric measures of market fragility (Samal et al., 2021)—rather than foregrounding the traversal algorithms themselves. The application of BFS, DFS and component detection therefore tends to remain implicit rather than explicit: these algorithms may underpin shortest-path, connectivity or related calculations, but they are less commonly treated as the primary analytical lens through which an equity correlation network is examined.

This gap is worth addressing for a few reasons. First, traversal algorithms offer a transparent, interpretable window into network structure. Unlike some higher-level metrics, the outputs of BFS and DFS—path lengths, spanning trees, articulation points, component memberships—have clear geometric meanings and can be interpreted directly without additional modelling assumptions. Second, an algorithmic approach that explicitly traces how the network can be traversed may capture aspects of market connectivity that summary statistics alone do not reveal. Third, from a computer science perspective, applying these algorithms to financial data provides a concrete case study in which classical algorithmic concepts are given empirical content. This thesis addresses this gap through an empirical application of graph traversal algorithms to stock correlation networks.

1.6 Python and NetworkX as Implementation Tools

The empirical analysis is implemented in Python, which has become the standard language for data science and computational finance. Its ecosystem of libraries—in particular NumPy, pandas, and Matplotlib—supports the full data pipeline from raw price data to network visualisation in a single, coherent environment (McKinney, 2010; Van Rossum and Drake, 2009).

For graph construction and analysis, the thesis relies on the NetworkX library (Hagberg et al., 2008). NetworkX provides native Python implementations of all the graph algorithms used in this project, including BFS, DFS, connected components, and MST construction. It integrates naturally with pandas DataFrames, which makes it straightforward to build a network from a correlation matrix and to attach node-level attributes such as sector labels and company names. Visualisation is handled through NetworkX’s drawing utilities in combination with Matplotlib.

The choice of Python and NetworkX also reflects a commitment to reproducibility. Both tools are freely available under open-source licences, and the code developed for this thesis can be run on any standard machine without proprietary dependencies. This matters in a research context: it means that the analysis can be verified, extended, or adapted by other researchers without access to specialised software (Peng, 2011).

1.7 Research Objectives

The thesis has three main objectives, which are pursued through the combination of empirical data analysis and graph-algorithmic methods.

The first objective is to construct stock correlation networks from historical daily return data for a sample of equities drawn from the S&P 500 index. This involves computing the pairwise Pearson correlation matrix, applying both threshold-based and MST filtering methods, and instantiating the resulting networks as weighted graph objects in NetworkX.

The second objective is to apply BFS, DFS, and connected component algorithms to the constructed networks and to analyse what their outputs reveal about the structural organisation of the equity market. Concretely, this means computing path length distributions, identifying articulation points, examining how component structure changes with the correlation threshold, and relating these findings to the sectoral composition of the stock universe.

The third objective is to discuss the results in economic terms. The aim here is not to develop a financial model or to conduct a formal portfolio optimisation exercise, but to assess whether the structural properties identified by graph traversal—clustering, connectivity, and bridging nodes—correspond to recognisable features of the equity market such as sector groupings, cross-sector connections, and diversification-relevant network

positions. This interpretive step is what connects the algorithmic analysis to the broader research question.

1.8 Research Methodology

The methodology is empirical and computational. It does not involve the derivation of new theoretical results or the development of new algorithms; rather, it applies established methods from both computer science and financial network analysis to a concrete dataset and interprets the results.

The dataset consists of daily adjusted closing prices for a selection of stocks from the S&P 500, covering a multi-year period that includes both relatively calm market conditions and periods of elevated volatility. Daily log-returns are computed from the price series, and the Pearson correlation matrix is estimated over the full sample window. Two filtering approaches are then applied to produce network representations: a threshold method, which retains edges above a series of correlation cutoffs, and the MST method of Mantegna (1999), which extracts the correlation backbone of the market.

The filtered networks are imported into NetworkX as weighted `Graph` objects, with node attributes including the sector classification of each stock. BFS is then applied from the configured source stock—AAPL in the reported analysis—to compute shortest-path distances across the network. DFS is used to explore traversal and spanning-tree structure, while articulation points and bridges are computed separately using NetworkX’s Tarjan-based routine. Connected component analysis is conducted across all threshold levels to characterise how the cohesion of the network changes with the filtering stringency.

The results are presented through a combination of descriptive statistics, network visualisations, and tables comparing structural properties across different threshold values and across subsets of the network defined by sector membership. The goal is clarity and interpretability rather than exhaustive enumeration of every possible graph metric.

1.9 Structure of the Thesis

The thesis is organised into five chapters. The present chapter has introduced the topic, defined the research gap, stated the objectives, and described the methodology.

Chapter 2 provides the theoretical background. It covers the relevant literature on stock correlation networks and financial network analysis, introduces the mathematical foundations of graph theory at the level needed for the empirical chapters, and describes the BFS, DFS, and connected component algorithms in more formal detail.

Chapter 3 presents the methodology and implementation. It describes the data collection and preprocessing pipeline, correlation and graph construction, network metrics, connected component analysis, the custom BFS and DFS implementations and their

validation, articulation points and bridges, and the Minimum Spanning Tree. It also introduces the exploratory investment framework and illustrative portfolio comparison used in the empirical analysis.

Chapter 4 presents the empirical analysis. It reports the results of applying BFS, DFS, and connected component algorithms to the constructed stock correlation networks, and interprets the structural findings in relation to the sectoral composition of the market and to the economic context of the sample period.

Chapter 5 concludes the thesis. It summarises the main findings, reflects on the limitations of the study, and suggests directions for future work. Two appendices follow the main text. Appendix A contains supplementary figures, while Appendix B provides reproducibility materials and selected core Python implementations. The complete source code is available in the accompanying GitHub repository, with a frozen copy of the thesis version also provided as `thesis_code.zip`.

Chapter 2

Theoretical Foundations

This chapter presents the background required for the empirical analysis in Chapters 3 and 4. It covers graph theory fundamentals, the key ideas from network science, the construction and interpretation of stock correlation networks, and the graph traversal algorithms at the centre of the study. A final section reflects on the practical relevance and limitations of the approach. Throughout, the focus is on concepts that are directly used later in the thesis; the goal is a clear working foundation rather than a comprehensive survey of the field.

2.1 Graph Theory Fundamentals

Graph theory provides the mathematical language for representing and analysing relational structures. Its origins lie in Euler’s 1736 solution to the Königsberg bridge problem, which established connectivity as a formal object of mathematical enquiry (Diestel, 2017). Today, graph theory is a foundational tool in computer science, operations research, and, increasingly, financial network analysis.

Formally, a graph G is defined as an ordered pair

$$G = (V, E),$$

where V is a finite non-empty set of vertices (nodes) and E is a set of edges, each connecting a pair of vertices. When edges have no directional meaning—as in a symmetric relationship like return correlation—the graph is called undirected. Directed graphs assign an orientation to each edge, making them appropriate for asymmetric relationships such as order flow or information causality (West, 2001). Since stock correlations are symmetric, all networks in this thesis are undirected.

In a weighted graph, each edge carries a numerical value representing the strength or cost of the connection. For stock correlation networks, weights are derived from the correlation or distance between pairs of return series. Unweighted graphs represent only

the presence or absence of an edge. Both forms are used here: the full correlation network is weighted, while threshold-filtered subgraphs are sometimes treated as unweighted once edges below the threshold have been removed (Mantegna, 1999).

A path in a graph is a sequence of distinct vertices $v_1, v_2, \dots, v_k$ where each consecutive pair is connected by an edge. The length of a path in an unweighted graph is the number of edges it contains. A cycle is a closed path that begins and ends at the same vertex. The shortest path between two vertices—the one with fewest edges—is one of the most practically useful concepts in network analysis. In a stock correlation network, shortest paths measure topological proximity: two stocks are “close” if they can be reached through a short chain of significant correlations. Computing shortest paths efficiently is the task of BFS for unweighted graphs and Dijkstra’s algorithm for weighted ones (Cormen et al., 2009).

A graph is connected if every pair of vertices is linked by at least one path. If some vertices cannot be reached from others, the graph is disconnected, and its maximal connected subgraphs are called connected components. In a threshold-filtered correlation network, stocks that share no sufficiently strong correlation with any other stock appear as isolated singleton components; the remaining stocks form one or more larger clusters. Figure 2.1 illustrates these concepts on a small example. How components form and dissolve as the threshold varies is a central part of the analysis in Chapter 4.

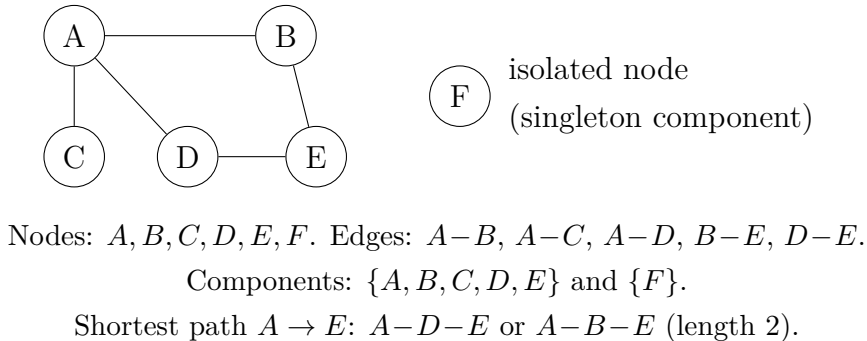


Figure 2.1: A simple undirected graph illustrating core graph-theoretic concepts used in this thesis. Nodes represent entities such as stocks; edges represent relationships such as significant correlations. Node F forms an isolated singleton component, analogous to a stock with no significant correlation to any other in the sample.

Two data structures are standard for representing graphs computationally. An adjacency matrix stores the full $N \times N$ array of edge weights, allowing $O(1)$ edge lookups at the cost of $O(N^2)$ space regardless of how many edges exist. An adjacency list stores, for each vertex, only the list of its neighbours, requiring $O(|V| + |E|)$ space. For the sparse graphs produced by threshold filtering, adjacency lists are considerably more efficient and are the internal representation used by NetworkX (Hagberg et al., 2008; Cormen et al., 2009). In this thesis, the correlation matrix is computed as a dense $N \times N$ structure, but the

filtered graph retains only a fraction of the possible edges. Table 2.1 summarises the key trade-offs.

Table 2.1: Adjacency Matrix vs. Adjacency List

Property	Adjacency Matrix	Adjacency List
Space complexity	$O(N^2)$ —full $N \times N$ array always allocated	$O(V + E)$ —proportional to edges present
Edge lookup	$O(1)$ —direct index access	$O(\text{degree})$ —scans neighbour list
Iterating over neighbours	$O(N)$ —must scan full row	$O(\text{degree})$ —directly available
Best suited for	Dense graphs (most pairs connected)	Sparse graphs (few edges relative to N^2)
Role in this thesis	Correlation matrix C stored as dense $N \times N$	Filtered graph stored as adjacency list in NetworkX

Comparison of adjacency matrix and adjacency list representations. In this thesis, the full correlation matrix is stored as a dense $N \times N$ structure; the filtered graph is represented internally as a sparse adjacency list by NetworkX (Cormen et al., 2009; Hagberg et al., 2008).

The degree of a vertex is the number of edges incident to it. In a weighted graph, the weighted degree (or strength) is the sum of the weights of incident edges. In a filtered stock correlation network, a high-degree node is an equity whose returns co-move significantly with many others—often a large-cap stock or one with broad exposure to systematic market factors. The degree distribution across all vertices gives a first characterisation of the network’s topology and helps distinguish hub-dominated configurations from more evenly spread ones (Barabási and Albert, 1999).

2.2 Network Science and Complex Systems

Network science emerged as a coherent field in the late 1990s, driven by the empirical discovery that real-world networks—social, biological, technological, and financial—share structural regularities absent from simple random graph models (Newman, 2010). Two papers in particular shaped early directions: Watts and Strogatz (1998) characterised the small-world property, and Barabási and Albert (1999) proposed a model for scale-free degree distributions. Both pointed to a common insight: despite the diversity of their components, complex systems tend to organise into networks with recognisable, recurring patterns.

A complex system is one where interactions among components produce collective behaviour that cannot be deduced from the properties of the individual parts (Barabási, 2016). Financial markets are a standard example: asset prices reflect the aggregated

decisions of many participants responding to common information, private signals, and each other’s behaviour. The resulting dynamics—volatility clustering, fat-tailed return distributions, contagion—arise from this interaction (Cont, 2001; Mantegna and Stanley, 2000). Network representations make part of this interaction structure explicit and tractable.

The small-world property refers to networks that combine short average path lengths with high local clustering (Watts and Strogatz, 1998). The clustering coefficient of a vertex v is defined as the fraction of pairs among v ’s neighbours that are themselves connected. Networks with both short global paths and dense local neighbourhoods are said to exhibit the small-world effect. In equity markets, Onnela et al. (2003) showed that stock correlation networks display this behaviour: even after filtering to retain only the strongest correlations, the average path length between stocks remains short.

Scale-free networks are characterised by a power-law degree distribution

$$P(k) \sim k^{-\gamma},$$

where a small number of high-degree hub nodes coexist with a large majority of low-degree nodes. This pattern, associated with preferential attachment dynamics, differs markedly from the narrow degree distribution of random graphs (Barabási and Albert, 1999). In equity networks, hub-like behaviour is observed for stocks with broad cross-sector exposure, such as large diversified conglomerates (Bonanno et al., 2004). Whether equity correlation networks are strictly scale-free is debated, but the presence of disproportionately well-connected nodes is a consistent empirical finding.

The average clustering coefficient—the mean of local clustering values across all vertices—captures the overall tendency towards local cohesion. High average clustering in a stock network reflects dense sector clusters: stocks in the same industry respond to shared drivers and therefore exhibit elevated pairwise correlations. This sectoral clustering is one of the most consistent findings in the empirical literature and provides the primary economic framework for interpreting the results in Chapter 4 (Mantegna, 1999; Bonanno et al., 2004).

2.3 Financial Networks and Stock Correlation Analysis

The term financial network refers broadly to graph representations of relationships among financial entities. Well-studied examples include interbank lending networks, where nodes are banks and edges represent bilateral loan obligations (Gai and Kapadia, 2010), and stock correlation networks, where nodes are equities and edges reflect the statistical co-movement of their returns. This thesis is concerned exclusively with the latter. Figure 2.3 at the

end of this section summarises the full analytical pipeline from raw data to structural interpretation.

The starting point is a panel of return time series. Given daily closing prices $P_{i,t}$ for stock i at time t , the logarithmic return is defined as

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right).$$

Log-returns are preferred to simple returns because they are additive over time and more closely approximate a normal distribution over short intervals (Campbell et al., 1997). For a universe of N stocks over T trading days, the dataset forms an $N \times T$ matrix of log-returns.

The linear dependence between the return series of stocks i and j is measured by the Pearson correlation coefficient, defined as

$$C_{ij} = \frac{\mathbb{E}[r_i r_j] - \mathbb{E}[r_i] \mathbb{E}[r_j]}{\sigma_i \sigma_j},$$

where σ_i and σ_j are the standard deviations of the respective return series and $\mathbb{E}[\cdot]$ denotes the sample mean. $C_{ij} \in [-1, 1]$: values near 1 indicate strong co-movement, values near -1 indicate opposing movements, and values near 0 suggest little linear relationship. Computing C_{ij} for all $N(N-1)/2$ pairs yields the symmetric $N \times N$ correlation matrix C , which is the primary input to network construction (Mantegna, 1999; Tumminello et al., 2010).

A key limitation of the Pearson coefficient is that it captures only linear dependence; non-linear relationships and tail co-movements are not reflected. The estimated matrix is also sensitive to the estimation window. A fixed window over the full sample period is used in this thesis, a common choice in exploratory stock-correlation-network studies (Onnela et al., 2003; Bonanno et al., 2004).

Since the full correlation matrix encodes $O(N^2)$ relationships, a filtering step is needed to produce a sparse, analytically tractable graph. Two complementary approaches are used.

The first is threshold filtering: an edge between stocks i and j is retained when

$$|C_{ij}| \geq \tau$$

for a chosen threshold $\tau \in [0, 1]$. By varying τ , one can study how the network changes as progressively strict correlation requirements are imposed. At low thresholds the graph is dense; at high thresholds only the strongest correlations survive and the graph fragments into separate components. This behaviour is directly analysed in Chapter 4.

The second approach is the Minimum Spanning Tree (MST), introduced to financial

networks by Mantegna (1999). The MST converts the correlation matrix into a distance metric

$$d_{ij} = \sqrt{2(1 - C_{ij})},$$

then extracts the spanning tree of minimum total distance—the set of $N - 1$ edges that connects all N stocks without forming any cycle. The MST is computed using Kruskal’s algorithm in $O(|E| \log |E|)$ time (Cormen et al., 2009). Mantegna (1999) showed that stocks on the same branch of the MST typically belong to the same economic sector, providing evidence that the MST reflects economically recognisable sectoral structure embedded in the correlation data. Figure 2.2 illustrates this transformation on a small example.

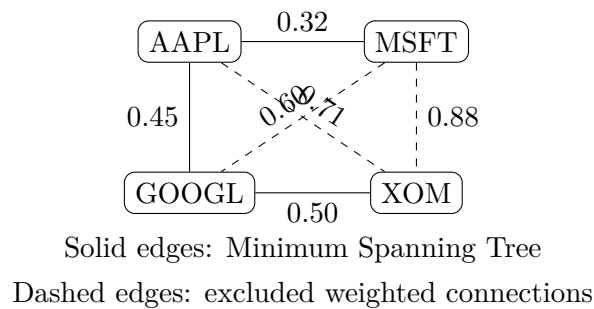


Figure 2.2: Illustrative transformation from a weighted correlation distance graph to its Minimum Spanning Tree. The MST retains exactly $N - 1$ edges, connecting all stocks while minimising total distance. Stocks in the same economic sector typically appear on adjacent branches (Mantegna, 1999).

A consistent finding in the literature is that equity correlations are not stationary: they tend to rise during market downturns as stocks fall together (Onnela et al., 2003). More recent evidence likewise documents substantial time variation in the structure of US equity networks across the Global Financial Crisis, the COVID-19 pandemic, and the Russia–Ukraine war period (Choi, 2023). While this thesis does not conduct a full time-series analysis, this background motivates using a sample period that spans varied market conditions. The resulting static network therefore summarises correlations estimated across heterogeneous market environments, rather than representing any single market regime.

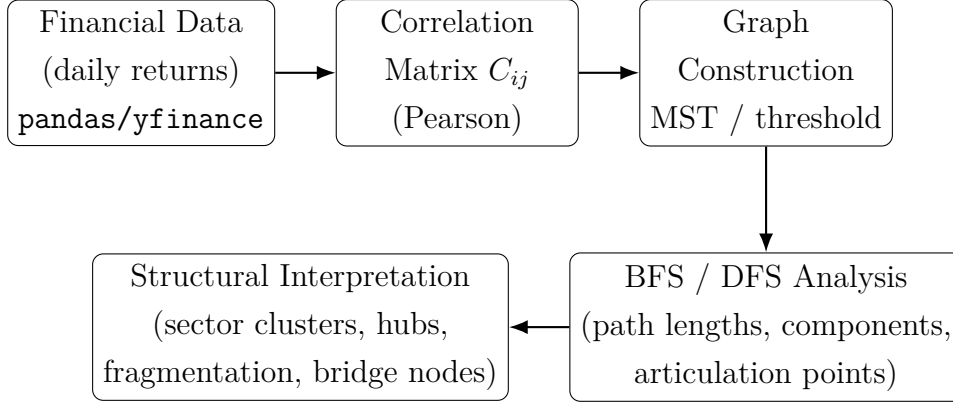
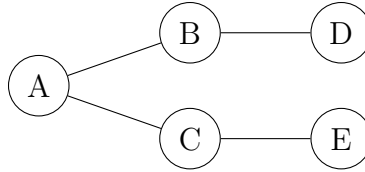


Figure 2.3: Analytical workflow of the thesis. Raw equity price data are transformed into log-returns and a Pearson correlation matrix using pandas. After MST or threshold-based filtering, the resulting graph is loaded into NetworkX and explored using BFS and DFS. Outputs are interpreted in the context of sector classifications and market conditions.

2.4 Graph Traversal Algorithms

Graph traversal algorithms systematically visit every vertex and edge of a graph. They are foundational in computer science: many graph problems—detecting components, computing paths, identifying critical nodes—can be solved through an appropriately instrumented traversal (Cormen et al., 2009; Sedgewick and Wayne, 2011). This thesis applies three techniques: Breadth-First Search, Depth-First Search, and connected component detection. Figure 2.4 shows how BFS and DFS produce different traversal orders on the same graph; Table 2.2 provides a structured comparison of their properties and financial relevance.



BFS visit order: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$.

DFS visit order: $A \rightarrow B \rightarrow D \rightarrow C \rightarrow E$.

Figure 2.4: BFS and DFS traversal of the same five-node graph from source node A . BFS explores layer by layer and computes shortest-path distances; DFS follows paths to their deepest point before backtracking, generating a spanning tree and enabling articulation-point detection. Both algorithms run in $O(|V| + |E|)$ time (Cormen et al., 2009).

Table 2.2: BFS vs. DFS: Properties and Financial Relevance

Property	BFS	DFS
Exploration strategy	Layer by layer (breadth-first)	Path to maximum depth, then backtrack
Data structure	Queue (FIFO)	Stack (LIFO) or recursion
Shortest-path guarantee	Yes—minimum edge-hop distances from source	No—path lengths not minimised
Articulation point detection	No	Yes—via Tarjan’s low-link values (Tarjan, 1972)
Spanning tree produced	Yes—BFS spanning tree	Yes—DFS spanning tree
Time complexity	$O(V + E)$	$O(V + E)$
Primary use in Chapter 4	BFS distances from AAPL, path lengths and diameter	Traversal coverage; articulation points and bridges via Tarjan
Financial interpretation	Measures integration and topological proximity	Identifies bridge stocks and hierarchical chains

Comparative summary of Breadth-First Search and Depth-First Search. The two algorithms run in identical time but differ in traversal order and the structural information they expose. Their outputs are complementary in the analysis of stock correlation networks.

Breadth-First Search (BFS). BFS begins at a source vertex s and explores the graph layer by layer: all vertices at distance 1 from s are visited first, then all at distance 2, and so on. The algorithm maintains a FIFO queue and records, for each vertex, the shortest-path distance from s and a predecessor pointer for path reconstruction. BFS runs in $O(|V| + |E|)$ time (Cormen et al., 2009).

Applied to a stock correlation network, BFS from a source stock computes the shortest-path distance to every other stock in the same connected component. These distances measure topological proximity: two stocks are “close” if they can be linked through a short chain of significant correlations. Averaging these over all pairs gives the network’s average path length, a global indicator of market integration. The diameter—the longest shortest path—provides an upper bound on separation between any two stocks. Networks with small diameters and short average path lengths are topologically compact, meaning that most stocks are connected through only a few surviving correlation links (Watts and Strogatz, 1998).

Depth-First Search (DFS). DFS starts from a source vertex and follows each path as deep as possible before backtracking. It uses a stack or recursion and also runs in $O(|V| + |E|)$ time (Cormen et al., 2009; Tarjan, 1972).

The most practically useful output of DFS in this context is the identification of articulation points—vertices whose removal would disconnect part of the graph. These can be found in a single $O(|V| + |E|)$ pass using Tarjan’s low-link algorithm (Tarjan, 1972). In a stock correlation network, an articulation point is an equity that occupies a structural intermediary position between parts of the graph. Its removal increases fragmentation in the estimated correlation network, revealing a topological role that would not be apparent from the stock’s individual return series alone. DFS also produces a spanning tree of the graph, giving a compact picture of which stocks serve as natural branches or entry points in the correlation hierarchy.

Connected Component Detection. A connected component is a maximal subgraph in which every pair of vertices is linked by at least one path. Components are identified by running BFS or DFS from each unvisited vertex; the total time is $O(|V| + |E|)$ (Cormen et al., 2009).

In this thesis, component analysis is applied across a range of threshold values τ to track how the correlation network fragments as filtering stringency increases. At low thresholds a single large component typically dominates; as τ rises, weakly correlated stocks detach and the large component breaks into smaller clusters. The rate and pattern of this fragmentation reveal how cohesive the market’s correlation backbone is, and which stocks remain tightly linked even at high correlation requirements. The size and sector composition of each component are among the key outputs reported in Chapter 4.

2.5 Structural Properties of Financial Networks

The traversal algorithms above produce outputs that characterise the financial network at both local and global levels. This section defines the specific indicators used in the empirical analysis and explains their financial meaning. The aim is not a comprehensive taxonomy of network metrics, but a clear account of the quantities that appear in Chapter 4.

Degree and Connectivity. The degree of a vertex is the number of edges incident to it. In a filtered stock correlation network, a high-degree stock co-moves significantly with many others—often a large-cap equity or one with broad exposure to systematic factors. The degree distribution gives a first summary of connectivity and helps distinguish hub-dominated topologies from more even ones (Barabási and Albert, 1999).

Average Path Length and Diameter. The average path length L is the mean of shortest-path distances over all pairs of vertices in the same connected component; the diameter D is the maximum. Both are computed via BFS and are sensitive to graph density. In financial terms, a short average path length implies that most stocks are linked

through few correlation hops—a sign of a tightly integrated market (Watts and Strogatz, 1998; Onnela et al., 2003).

Clustering Coefficient. The local clustering coefficient of vertex v is the fraction of pairs among v ’s neighbours that are also connected to one another. High local clustering signals a dense sector cluster where most peers also correlate with each other. The average clustering coefficient—the mean across all vertices—summarises this tendency across the whole network (Watts and Strogatz, 1998).

Hubs and Articulation Points. Hubs are high-degree nodes that disproportionately hold the network together. Articulation points, identified through DFS-based low-link methods such as Tarjan’s algorithm, are vertices whose removal increases the number of connected components—not all hubs are articulation points, and vice versa. In a correlation network, an articulation point is a stock whose removal increases the number of connected components. Its significance is therefore structural: the stock occupies a uniquely important position in the connectivity of the estimated network, although this does not by itself imply greater financial risk or superior investment relevance.

Fragmentation and Component Sizes. The fraction of nodes in the largest connected component—the “giant component”—measures network cohesion. When this fraction is close to 1 the market is tightly integrated; when it is small, the market fragments into poorly correlated clusters. Monitoring giant component size across threshold values τ provides a picture of how the correlation backbone is organised (Onnela et al., 2003).

Sectoral Clustering. Nodes in a stock correlation network carry economic metadata, most importantly industry sector. Comparing connected-component structure and Minimum Spanning Tree branches with known sector classifications provides a descriptive check on whether the network representation reflects recognised sectoral patterns (Mantegna, 1999; Bonanno et al., 2004).

2.6 Graph Algorithms in Quantitative Finance: Relevance and Limitations

Having established the theoretical framework, it is worth briefly reflecting on what a graph-algorithmic approach can and cannot deliver in a financial context. This helps set realistic expectations for the empirical chapters, which are exploratory in nature rather than aimed at predictive modelling or parameter estimation.

The main contribution of graph-based methods to financial analysis is making the relational organisation of markets explicit and computable. A correlation matrix containing

hundreds of entries is difficult to interpret directly; representing it as a graph and exploring it algorithmically converts that information into a smaller set of interpretable indicators—path lengths, component memberships, articulation points—that can be described and compared. This is particularly valuable in exploratory work, where the goal is to understand how the market is organised rather than to fit a model (Tumminello et al., 2010).

From a portfolio diversification standpoint, network analysis complements classical mean-variance optimisation by identifying which stocks are most peripheral in the correlation graph and therefore potentially most diversifying. Pozzi et al. (2013) showed that peripheral stocks in the MST tend to provide better out-of-sample diversification than central ones; Peralta and Zareei (2016) formalised this into a portfolio selection procedure, and Clemente et al. (2022) more recently embedded correlation-network structure directly into risk-based portfolio allocation models. This thesis does not implement a formal mean-variance portfolio optimisation model. Instead, the structural findings it generates—which stocks are hubs, which are peripheral, and how components align with sectors—are used later in a limited exploratory diversification comparison.

Network analysis also provides a lens for thinking about market-wide interconnectedness. A market in which most stocks are connected through short correlation chains exhibits a compact structure of observed co-movement. BFS path lengths and component sizes provide coarse but informative measures of this topological interconnectedness. It should be noted, however, that stock correlation networks measure statistical co-movement, not direct financial obligations. Conclusions about systemic risk in the strict sense require different data and models (Gai and Kapadia, 2010; Battiston et al., 2012). In this thesis, interconnectedness is used descriptively to refer to the degree of observed co-movement across the equity market.

Several limitations of the approach are worth acknowledging. First, the Pearson correlation captures only linear dependence and is sensitive to the estimation window and to outliers. Second, the choice of filtering threshold τ involves a degree of arbitrariness—which is why this thesis analyses the network across a range of values rather than a single fixed one. Third, a static network estimated from one window does not capture the time-varying nature of financial correlations; a rolling-window analysis would yield richer temporal information but would substantially increase the scope of the study and is left as a direction for future work.

These limitations define the scope within which the empirical results should be interpreted. Within that scope, BFS, DFS, and connected component detection provide a transparent and reproducible way to examine how equity markets are organised at the level of pairwise return correlations. The following chapters put this framework into practice.

Chapter 3

Methodology and Implementation

This chapter answers a single question: how was the empirical analysis implemented? It describes how the price data was obtained and cleaned, how the correlation matrix was turned into graphs, and how each graph-theoretic technique—connected components, Breadth-First Search, Depth-First Search, articulation points and bridges, and the Minimum Spanning Tree—was implemented and validated. It does not interpret what the analysis found: the substantive discussion of centrality, clustering, structural fragility and portfolio behaviour on the real-data network is left to Chapter 4. Where a real-run number is given here, it is included only to confirm that the pipeline executed as specified, not as an empirical finding in its own right.

Notation used throughout this chapter: N denotes the number of stocks in the universe (47 unless stated otherwise); C_{ij} the Pearson correlation between stocks i and j ; τ (“tau”) the correlation threshold applied to filter an edge; d_{ij} the Mantegna distance derived from C_{ij} ; and $G = (V, E)$ an undirected graph whose edges store the signed correlation C_{ij} , its absolute value $|C_{ij}|$, and the corresponding Mantegna distance d_{ij} . The complete pipeline is implemented in Python (Van Rossum and Drake, 2009).

3.1 Overview of the Empirical Pipeline

The implementation is organised around a single configuration module (`config.py`), an orchestrating entry point (`main.py`), and a set of supporting modules under `src/`, each responsible for a distinct stage of the empirical pipeline. Every parameter that a reader of the thesis might reasonably want to change—the sample period, the stock universe, the correlation thresholds, the BFS source stock—is declared once in `config.py` and imported everywhere else, so that the experiment can be reproduced, or varied, by editing a single file. `main.py` then executes the pipeline stages in sequence: data collection, preprocessing, correlation estimation, graph construction, network metrics, connected components, Breadth-First Search, Depth-First Search, the Minimum Spanning Tree, figure generation, the exploratory investment framework, the illustrative portfolio comparison

and a consolidated summary report. Figure 3.1 traces this sequence at the level relevant to a methodology chapter; the exported artefacts it produces are interpreted in Chapter 4.

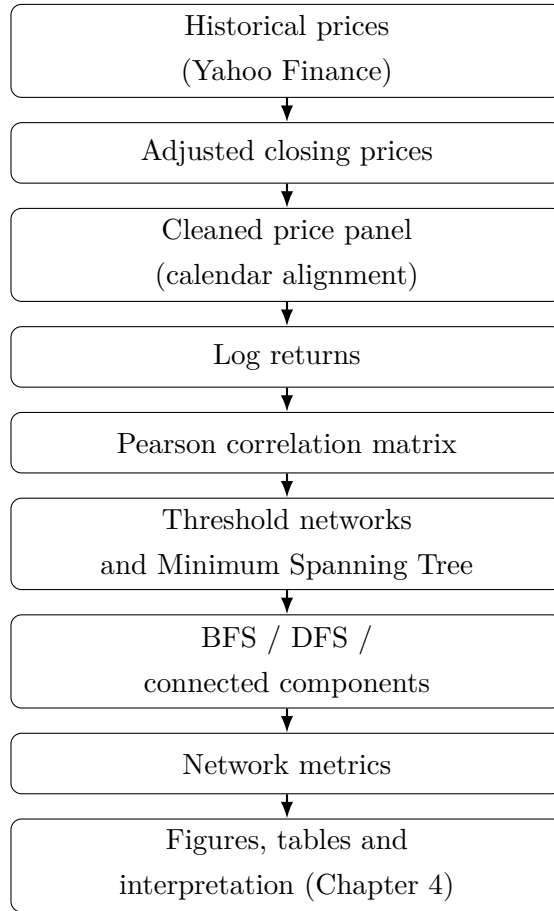


Figure 3.1: Methodological workflow of the empirical pipeline.

3.2 Data Collection

Daily adjusted closing prices for the 47-ticker universe are downloaded from Yahoo Finance through the `yfinance` library (`src/data_collection.py`). Adjusted, rather than raw, closing prices are used because a raw close jumps mechanically on an ex-dividend date or a stock split even though no economic gain or loss occurs; such artificial jumps would appear as large returns and would contaminate every correlation that involves the affected stock. This is the standard input convention in empirical finance (Campbell et al., 1997) and in the correlation-network literature specifically (Mantegna, 1999; Onnela et al., 2003).

The download routine retries a failed request automatically and normalises the different table layouts returned by `yfinance` into a single dates-by-tickers panel; a ticker that cannot be downloaded is logged and excluded rather than allowed to abort the run.

A second, distinct concern is provenance. The pipeline also supports a `-synthetic` self-test mode that simulates prices so that the code can be exercised offline; its output is

written to a separate directory and marked as simulated, so that a self-test can never be mistaken for, or silently mixed with, the real results used in this thesis. Every run also writes a short data-source report stating explicitly whether real Yahoo Finance data or simulated data was used. The run that grounds this chapter downloaded all 47 requested tickers from Yahoo Finance with no failures.

The 47 tickers are drawn from ten GICS sectors, listed in Table 3.1, with three to five names per sector by deliberate design. The universe is sector-balanced rather than capitalisation-weighted, so that no single sector’s size mechanically dominates the sectoral-clustering evidence examined in Chapter 4. The resulting universe is therefore a purposive research sample designed to provide broad sector coverage rather than a statistically representative sample of the entire S&P 500. The empirical conclusions are interpreted within this selected universe and are not generalised mechanically to all index constituents.

Table 3.1: Stock universe by sector (`config.STOCKS`).

Sector	Tickers	No. of stocks
Technology	AAPL, MSFT, NVDA, ADBE, CRM	5
Financials	JPM, BAC, GS, MS, WFC	5
Healthcare	JNJ, PFE, MRK, UNH, ABBV	5
Energy	XOM, CVX, COP, SLB, EOG	5
Consumer Staples	PG, KO, PEP, WMT, COST	5
Consumer Discretionary	AMZN, HD, MCD, NKE, SBUX	5
Industrials	BA, CAT, GE, HON, UNP	5
Communication Services	GOOGL, META, DIS, VZ, T	5
Utilities	NEE, DUK, SO, AEP	4
Real Estate	AMT, PLD, SPG	3
Total	47 tickers across 10 GICS sectors	47

3.3 Data Cleaning and Calendar Alignment

A Pearson correlation is defined only for two series measured at the same set of dates. If the 47 price series were compared over slightly different calendars, pandas (McKinney, 2010) would silently align each pair on its own overlapping dates, so every entry of the correlation matrix would be estimated on a different sample. `src/preprocessing.py` therefore enforces one common trading calendar for the whole universe before any correlation is computed, through a sequence designed to go from least to most destructive:

- sort the panel by date and drop duplicate index entries;
- drop any ticker that is empty over the whole window (a failed download);
- drop any ticker missing more than 5% of the common calendar (`config.MAX_MISSING_FRACTION`);
- forward-fill remaining gaps, but only up to 5 consecutive trading days (`config.MAX_FORWARD_FILL_DAYS`): a short gap such as a local holiday is carried forward, while a longer gap is left as missing rather than fabricated;
- drop any row still containing a missing value, which preserves the alignment property exactly;
- verify that at least 250 observations remain (`config.MIN_OBSERVATIONS`); the pipeline raises an explicit error rather than continuing on too short a sample.

Short gaps are forward-filled to preserve a common trading calendar while avoiding unnecessary loss of observations. The fill is capped at five consecutive trading days because carrying a price forward produces a return of exactly zero; long runs of such artificial zero returns would reduce estimated volatility and could bias correlations toward zero. Tickers with poor coverage are therefore dropped rather than extensively patched. On the real-data run, the full 47-stock universe survived this stage with no ticker removed, yielding 1,759 aligned daily return observations over 2019–2025.

3.4 Computation of Logarithmic Returns

Daily returns are computed as logarithmic returns,

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right),$$

rather than simple percentage returns, for reasons standard in empirical finance (Campbell et al., 1997). First, log returns are additive over time: the cumulative log return over several trading days is simply the sum of the individual daily log returns, which simplifies every downstream time aggregation. Second, the resulting series is approximately stationary, which the price level itself is not; correlating price levels would mostly capture a shared long-run trend rather than genuine daily co-movement.

A further, purely arithmetic property motivates the choice. A positive log return exactly offset by an equal-magnitude negative log return returns the price to its original level, because the two logarithms cancel in the sum that defines the cumulative return. Simple percentage returns do not have this property: a +10% move followed by a −10%

move leaves the price 1% below its starting level, since $1.10 \times 0.90 = 0.99$. For the small daily moves typical of liquid equities the two definitions are nevertheless numerically close, because $\ln(1 + x) \approx x$ for small x , so the distinction matters more as a modelling convention than as a source of materially different daily values.

A per-stock descriptive-statistics table (mean, standard deviation, skewness, excess kurtosis, and their annualised counterparts) is also exported. Documenting the well-known non-normality of daily equity returns is what justifies treating the Pearson correlation, in the next section, as a measure of linear dependence only—a limitation stated explicitly in Section 3.17. On the real run this stage produced 1,759 daily log-return observations for all 47 stocks.

3.5 Estimation of the Pearson Correlation Matrix

The $N \times N$ matrix of pairwise correlations is estimated from the cleaned return panel using pandas (McKinney, 2010),

$$C_{ij} = \frac{\text{cov}(r_i, r_j)}{\sigma_i \sigma_j}, \quad C_{ij} \in [-1, 1].$$

Because the panel is aligned to a single calendar, every coefficient is estimated on exactly the same set of dates. The matrix is explicitly symmetrised and its diagonal reset to exactly one, which removes floating-point noise that several downstream graph routines are sensitive to. For 47 stocks the matrix contains $N(N - 1)/2 = 1,081$ distinct pairs. A companion routine aggregates these into a sector-by-sector average-correlation matrix, computed before any graph is built, so that Chapter 4 can relate the network-level clustering result back to the raw correlation structure that produced it.

3.6 From Correlation Matrix to Graph

The correlation matrix is turned into an undirected graph $G = (V, E)$, where V is the set of 47 stocks—each carrying a sector and company-name attribute—and E is a set of unordered stock pairs. Each edge stores several related quantities: the signed Pearson correlation C_{ij} , its absolute value $|C_{ij}|$, and the Mantegna distance d_{ij} . In the NetworkX representation, the generic `weight` attribute is set to $|C_{ij}|$, while the signed value is retained separately as `correlation`. This allows different downstream measures to use the representation appropriate to their interpretation. This representation follows the standard construction used in the econophysics literature on financial correlation networks (Mantegna, 1999; Onnela et al., 2003; Tumminello et al., 2010).

Two families of such graphs are built in `src/graph_construction.py`. The complete correlation graph keeps all 1,081 pairs; it preserves the full information of the correlation

matrix but is not, on its own, an informative picture, since every node is adjacent to every other by construction. It is retained as the reference object from which every filtered graph, and the Minimum Spanning Tree of Section 3.13, is derived.

Every edge of the complete graph also stores the Mantegna distance (Mantegna, 1999),

$$d_{ij} = \sqrt{2(1 - C_{ij})},$$

which converts the similarity measured by a correlation into a proper metric: d is non-negative, equals zero only for perfectly correlated stocks, is symmetric, and satisfies the triangle inequality. This is what licenses its later use as an edge weight in the Minimum Spanning Tree.

3.7 Threshold-Filtered Graph Construction

A dense correlation matrix is not a readable network, so the standard approach—adopted here—is to keep only the pairs whose dependence exceeds a cut-off τ . An edge (i, j) survives when

$$|C_{ij}| \geq \tau.$$

The absolute-value rule treats a strongly negative relationship as equally informative as a strongly positive one, since for risk and diversification purposes a pair of stocks that systematically moves in opposite directions is strongly related, not unrelated. A “positive-only” mode is available in the configuration for robustness checks but is not the specification used for the reported results; whether it would matter empirically on this sample is a question for Chapter 4. Nodes are always added to the graph before any edge, so that a stock with no surviving correlation still appears as an isolated node rather than being silently dropped.

This convention is deliberately different from the MST construction. In the threshold graphs, $|C_{ij}|$ measures the strength of statistical dependence regardless of sign; in the MST, the signed correlation enters $d_{ij} = \sqrt{2(1 - C_{ij})}$, so strongly negative correlations correspond to large distances. The two graph families therefore answer different questions: threshold graphs retain strong dependence of either sign, whereas the MST traces the backbone of strongest positive co-movement.

The threshold grid used throughout is

$$\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\},$$

with $\tau = 0.5$ designated the main threshold for the detailed single-network analyses of Sections 3.9–3.12. The threshold is treated as a genuine modelling parameter: how strongly the network’s structure depends on it is examined empirically in Chapter 4. Table 3.2

summarises this and the other configuration parameters used in the reported run.

Table 3.2: Configuration parameters of the empirical pipeline (`config.py`).

Parameter	Value used in the empirical run
Sample period	2019-01-01 to 2026-01-01 (first/last trading day: 2019-01-02 / 2025-12-31)
Stock universe	47 tickers across 10 GICS sectors, 3–5 names per sector (sector-balanced, not capitalisation-weighted)
Return definition	Daily logarithmic return, $r_t = \ln(P_t/P_{t-1})$, computed on adjusted close
Max. missing fraction	5% of trading days (a ticker exceeding this is dropped)
Max. forward-fill	5 consecutive trading days
Min. observations	250 return observations required to keep a ticker
Correlation thresholds (τ)	0.3, 0.4, 0.5, 0.6, 0.7
Main threshold	$\tau = 0.5$ (used for the detailed BFS, DFS and component analysis)
Edge-filter rule	“absolute”: an edge survives when $ C_{ij} \geq \tau$
BFS source stock	AAPL (falls back to the highest-degree node if missing or isolated)
MST distance	Mantegna: $d_{ij} = \sqrt{2(1 - C_{ij})}$; signed correlation
Centrality quantiles	Central $\geq$ 80th percentile; peripheral $\leq$ 25th percentile of composite centrality
Illustrative portfolio size	10 stocks per portfolio
Random seed	42 (layout algorithms and the offline self-test only)

3.8 Network Metrics

`src/metrics.py` computes one row of node-level statistics per stock on every threshold graph: degree and strength, with strength defined using $|C_{ij}|$; degree, betweenness and closeness centrality; eigenvector centrality using $|C_{ij}|$ as the edge weight; and the clustering coefficient in both unweighted and $|C_{ij}|$ -weighted form. Betweenness is computed both on the unweighted topology and with the Mantegna distance as edge weight, so that a path through strongly positively correlated stocks counts as shorter in the distance-weighted variant. Closeness applies the Wasserman–Faust correction for disconnected graphs (Wasserman and Faust, 1994), which evaluates reachability within a node’s connected component and rescales the resulting score by the fraction of the full network that is

reachable. This makes closeness values more meaningful when comparing nodes belonging to components of different sizes; isolated nodes receive a closeness value of zero.

Eigenvector centrality is computed with a numerically robust routine that degrades gracefully on the disconnected, high-threshold graphs where the standard power iteration does not reliably converge. No single centrality measure is treated as authoritative—degree, betweenness, closeness and eigenvector centrality each capture a different notion of importance—so a composite centrality is also computed as the mean of the four measures after min–max normalisation. The equal weighting of the four measures, as well as the 80th- and 25th-percentile cut-offs later used to label stocks as central or peripheral, are heuristic design choices intended to create interpretable descriptive groups; they are not theoretically derived categories and do not replace the individual centrality measures. At the graph level, the module reports density, average degree, average clustering, transitivity, the count and size of connected components, sector and degree assortativity, and, restricted to the giant component, the average shortest path length and diameter. How these measures differ across stocks on the real-data network is reported in Chapter 4.

3.9 Connected Component Analysis

A connected component is a maximal set of stocks joined, directly or through intermediaries, by surviving edges. Components answer a structural question that the threshold alone cannot: does the market break into separate blocks of co-moving stocks once weak links are removed, or does it remain one cohesive whole? Each stock is labelled with a component identifier (0 for the largest component, the “giant component”), together with the most frequent sector in that component and its share. These labels are descriptive only: when several sectors have the same largest share, no single sector is interpreted as substantively dominant. The sector composition of each component is therefore considered alongside the component size rather than from the label alone. At lower thresholds, the universe is more likely to form one large connected component, reflecting the common market factor, while at higher thresholds the network is expected to fragment. The number, size and sector composition of the components obtained on the real-data network are reported in Chapter 4.

3.10 Breadth-First Search

Breadth-First Search is implemented from first principles in `src/traversal_analysis.py` (`custom_bfs`), rather than called directly from NetworkX (Hagberg et al., 2008), so that the algorithm’s logic can be presented and discussed on its own terms. It maintains a FIFO queue of discovered-but-unexplored nodes; because every edge of a threshold graph is unweighted, the order in which nodes are dequeued is exactly the order of increasing

distance from the source, which is why BFS also solves the shortest-path problem here. A deterministic tie-breaking rule (neighbours are visited in a fixed order) makes the traversal reproducible. The algorithm runs in $O(|V| + |E|)$ time and $O(|V|)$ space (Cormen et al., 2009).

The financial reading of the output is direct: the BFS distance from a source stock is the minimum number of “strong correlation” links needed to reach another stock, and is therefore a measure of topological proximity inside the estimated dependence structure rather than a forecast of anything. The source stock is AAPL by default (`config.SOURCE_STOCK`); if the configured source is absent from the final universe or isolated at the chosen threshold, the pipeline substitutes the highest-degree node and records the substitution. The specific reachability pattern obtained on the real-data network—which stocks are reached at each distance from AAPL, and which, if any, are unreachable—is reported and interpreted in Chapter 4.

3.11 Depth-First Search, Articulation Points and Bridges

Depth-First Search is likewise implemented from first principles (`custom_dfs`) using an explicit stack rather than recursion, with the same deterministic tie-breaking convention adopted for BFS. Restarting this single-source DFS from every not-yet-visited node produces a DFS forest whose individual trees are exactly the graph’s connected components—the classical way of recovering components from a traversal (Cormen et al., 2009).

On the same threshold graph, articulation points and bridges are computed separately using NetworkX’s Tarjan-based reference routines, which implement linear-time graph algorithms (Tarjan, 1972). An articulation point is a stock whose removal increases the number of connected components. Rather than reporting this as a bare property, the corresponding analysis records how the component structure changes after the stock is removed. The substantive interpretation in Chapter 4 focuses on the articulation-point status and the resulting change in connectivity rather than on an auxiliary list of individual “detached” stocks. A bridge is a single edge whose removal disconnects the network, reported together with the size of the two sides it holds together. A companion routine counts both quantities across the whole threshold grid, since a dense, low-threshold graph has so many redundant paths that removing any single stock leaves the rest connected—articulation points and bridges appear only once the network becomes sparse enough. As with every traversal-derived result, the exported tables carry the explicit caveat that an articulation point is a structural property of one estimated network over one sample, not a quality judgement about the company. The specific articulation points and bridges identified on the real-data network, and their financial interpretation, are presented in Chapter 4.

3.12 Validation of the Custom Traversal Implementations

Because the BFS and DFS routines above are written from scratch rather than called from a library, their outputs are checked against NetworkX reference routines immediately after execution. For BFS the validation is exact: shortest-path distances are canonical, so the custom distances are compared node-by-node against `nx.single_source_shortest_path_length`; any mismatch would indicate an implementation error. For DFS, the exact pre-order is not used as the primary validation target because a valid DFS order depends on the order in which neighbours are examined. The custom routine is therefore checked for complete and non-duplicated coverage of the reachable nodes, and the connected components recovered by the full DFS forest are compared directly with `nx.connected_components`. These checks validate reachability and component recovery, but they are deliberately not presented as a proof that every possible DFS ordering property has been independently verified.

Table 3.3 lists every check performed and its outcome on the real-data network at the main threshold ($\tau = 0.5$). The BFS result is stated explicitly in the pipeline’s own summary report (“Validation of the custom BFS against NetworkX: all checks passed”). Articulation points and bridges are computed separately with NetworkX’s Tarjan-based reference routines, as described in Section 3.11; they are therefore not treated as outputs whose correctness depends on the custom DFS validation.

Table 3.3: Validation checks implemented for the custom BFS and DFS routines, and their outcome on the real-data network.

Algorithm	Check	What it verifies	Outcome
BFS	Reached set	Same reachable component as NetworkX	PASS
BFS	Distances	Same shortest-path distances as NetworkX	PASS
BFS	Layer order	Nodes processed in non-decreasing distance order	PASS
DFS	Visited set	Same visited nodes as NetworkX	PASS
DFS	No repeated visits	No node is processed twice	PASS
DFS	Reachability coverage	All reachable nodes are visited exactly once	PASS
DFS forest	Components	Same connected components as NetworkX	PASS

3.13 Minimum Spanning Tree

The Minimum Spanning Tree of the Mantegna distance graph is built with NetworkX’s implementation of Kruskal’s algorithm (Kruskal, 1956): edges are sorted by increasing distance and added greedily whenever they join two components that are not yet connected. Because the Mantegna transformation is monotonically decreasing in the correlation, minimising total distance is equivalent to keeping the strongest correlations that do not create a cycle. With the signed correlation entering the distance formula, more negatively correlated pairs are assigned larger distances, with the maximum distance $d_{ij} = 2$ occurring when $C_{ij} = -1$. The tree therefore preferentially retains pairs with stronger positive co-movement—the convention adopted throughout the econophysics literature on financial correlation networks (Mantegna, 1999; Tumminello et al., 2010).

The tree keeps exactly $N - 1 = 46$ of the 1,081 possible correlations—the smallest set of links that still connects every stock—without requiring any threshold. It is therefore a threshold-free complement to the threshold-filtered graphs of Section 3.7. Node-level output includes the tree degree, tree betweenness and closeness, eccentricity, and depth from the tree’s highest-degree hub; sector-level output reports, for every sector, how many of its members’ tree links stay internal versus lead outside the sector. The identity of the tree’s hub, the proportion of leaf stocks, and the extent to which its links follow sector lines are reported and interpreted in Chapter 4.

3.14 Exploratory Investment Framework

`src/interpretation.py` attaches four deliberately non-exclusive descriptive flags to every stock, built from the network quantities computed above: `is_central` (composite centrality at or above the 80th percentile); `is_peripheral` (composite centrality at or below the 25th percentile, or a leaf of the Minimum Spanning Tree); `is_bridge` (the stock is an articulation point); and a cluster label naming the connected component the stock belongs to. A single headline network role is then assigned, together with one auto-generated sentence per stock that states its position in descriptive terms, without characterising the company itself or its expected return.

Every table and log entry this module produces carries the following disclaimer verbatim:

“This framework is exploratory and supports diversification and risk interpretation. It does not constitute investment advice and does not predict future returns.”

Centralising the wording in a single configuration constant guarantees that it appears identically wherever the framework’s output is used. The distribution of stocks across these descriptive roles on the real-data network is reported in Chapter 4.

3.15 Illustrative Portfolio Comparison

`src/portfolio.py` adds a purely illustrative comparison between a sector-diversified portfolio and a network-diversified portfolio. Portfolio A selects, from each of the ten sectors, the single stock with the highest average correlation to the rest of its own sector—a deterministic, sector-based and non-topological rule that makes Portfolio A a benchmark distinct from the graph-topology-based selection used for Portfolio B. Portfolio B is built by a transparent greedy rule. Each candidate receives a peripherality score

$$s_i = 0.5(1 - \tilde{c}_i) + 0.3(1 - \tilde{k}_i) + 0.2\ell_i,$$

where $\tilde{c}_i$ is the min–max-normalised composite centrality, $\tilde{k}_i$ is the normalised degree, and ℓ_i equals one when stock i is a leaf of the Minimum Spanning Tree and zero otherwise. The stock with the highest peripherality score seeds the portfolio.

For each subsequent selection, stocks belonging to a connected component not yet represented in the portfolio are prioritised whenever such candidates remain. Within the eligible candidate set, the algorithm selects the stock that minimises its maximum absolute correlation with the stocks already included,

$$\min_i \max_{j \in P} |C_{ij}|,$$

where P denotes the set of previously selected stocks. Remaining ties are resolved using the peripherality score. Once all available components are represented, selection continues according to the same minimum-maximum-correlation criterion. Using the maximum rather than the average pairwise correlation discourages the rule from accepting one highly correlated pair in exchange for a favourable average. Portfolio C is the full 47-stock universe, equally weighted, as an unconstrained benchmark.

Every portfolio is equally weighted and rebalanced daily; portfolio returns are aggregated from simple, not logarithmic, returns, since a portfolio’s return is the weighted average of its constituents’ simple returns and log returns are not additive across assets. Evaluation reports annualised return and volatility, the Sharpe ratio, maximum drawdown and average pairwise correlation, following standard practice in empirical asset-pricing work (Campbell et al., 1997). For this illustrative comparison, the Sharpe ratio is calculated using a zero risk-free rate, consistent with the assumption specified in the implementation.

The comparison is illustrative rather than evaluative: the portfolios are formed and evaluated on the same historical sample, so the exercise is entirely in-sample and no out-of-sample test is performed; transaction costs, taxes and liquidity constraints are not modelled; and the resulting portfolio illustration provides a limited, in-sample demonstration of how network-based information may support diversification analysis, not a validated investment strategy. The results are not investment advice and do not predict future performance.

The risk, return and diversification statistics obtained for the three portfolios on the real-data run are reported and discussed in Chapter 4.

3.16 Large-Network Extension

The analysis in Sections 3.1–3.15 is built around a purposive, sector-balanced universe of 47 stocks. As a supplementary robustness and scalability check, the same empirical framework is also applied to a substantially larger stock universe to examine whether the structural and sectoral patterns observed in the main analysis remain visible at a different scale. The extension does not replace the 47-stock experiment, which remains the primary controlled empirical study of the thesis. Instead, it reuses the core correlation-estimation, graph-construction and network-analysis routines of the original pipeline, while adding separate routines for large-universe execution, sector-alignment analysis and dedicated network exports. The original analysis and its reported outputs are therefore preserved unchanged. The substantive comparison between the two network sizes is presented in Chapter 4; the purpose of this section is to document how the larger experiment was constructed.

The extension starts from the latest available point-in-time S&P 500 membership snapshot not later than the end of the sample period. This snapshot is dated 22 December 2025 and contains 503 constituents distributed across 11 GICS sectors. Because the universe is defined according to index membership near the end of the sample rather than reconstructed dynamically throughout 2019–2025, the analysis is conditional on end-of-period membership. It should therefore not be interpreted as a survivorship-bias-free reconstruction of the S&P 500 over the full historical window.

The membership and sector metadata were obtained from a public, version-controlled historical S&P 500 constituent dataset and frozen as static reference files in the project repository.¹ For GICS metadata, the extension uses a snapshot dated 16 November 2025 as the primary source. This provides classifications for 498 of the 503 constituents. A later snapshot dated 17 January 2026 is used only as a fallback for the five constituents that are present in the December membership snapshot but absent from the earlier metadata file. No missing sector classification is inferred or assigned manually.

Of the 503 requested securities, historical adjusted closing prices for 495 were successfully downloaded from Yahoo Finance. Eight tickers were unavailable from the data source. The same data-quality rule used in the main analysis (Section 3.3) was then applied: a further 21 securities were excluded because more than 5% of their observations were missing in the downloaded price panel. The resulting large-network universe therefore contains 474

¹Historical constituent data were obtained from the public GitHub repository `fja05680/sp500` (<https://github.com/fja05680/sp500>). The project records the source revision used to construct the frozen reference files in `sp500_constituents_2025_12_22_provenance.json`.

stocks. Table 3.4 summarises the main configuration and sample construction.

Table 3.4: Configuration and sample construction of the large-network extension.

Item	Value
Constituents requested	503 tickers across 11 GICS sectors
Membership snapshot date	2025-12-22
Downloaded / unavailable	495 downloaded; 8 unavailable from Yahoo Finance
Excluded ($> 5\%$ missing prices)	21 tickers, using the same rule as Section 3.3
Final large-network universe	474 stocks
Sample period	2019-01-01 to 2026-01-01 (first/last observed trading day: 2019-01-02 / 2025-12-31)
Aligned common calendar	2019-03-20 to 2025-12-31
Aligned trading days	1,707
Log-return observations	1,706
Correlation thresholds (τ)	0.3, 0.4, 0.5, 0.6, 0.7
Main threshold	$\tau = 0.5$
Edge-filter rule	$ C_{ij} \geq \tau$, as in Section 3.7
GICS metadata, primary snap- shot	2025-11-16 (498 constituents)
GICS metadata, fallback snap- shot	2026-01-17 (5 constituents only)

The requested historical interval, price definition and preprocessing rules are kept as close as possible to those of the main experiment. Daily adjusted closing prices are used, short gaps are forward-filled for at most five consecutive trading days, and logarithmic returns are computed exactly as described in Section 3.4. After the missing-data filter and permitted forward fill have been applied, complete-case alignment removes every remaining date on which at least one of the 474 retained price series still contains a missing observation. Because this condition is imposed simultaneously on a much larger panel than in the 47-stock analysis, the resulting common calendar begins on 20 March 2019 rather than 2 January 2019. It contains 1,707 aligned price dates and therefore 1,706 daily log-return observations through 31 December 2025.

Pearson correlation is then estimated on the aligned panel using the same procedure as in Section 3.5. With $N = 474$, the resulting correlation matrix contains

$$\frac{N(N-1)}{2} = \frac{474 \times 473}{2} = 112,101$$

distinct stock pairs. The same threshold grid,

$$\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\},$$

and the same absolute-correlation edge rule,

$$|C_{ij}| \geq \tau,$$

are retained, with $\tau = 0.5$ again used as the main threshold. The extension reports the same graph-level quantities needed for comparison across thresholds, including the number of nodes and edges, density, average degree, average clustering, connected-component structure, giant-component size and isolated nodes. Sector assortativity and same-sector edge shares are added as descriptive measures once the network topology has already been constructed.

GICS labels play no role in determining whether an edge exists. Network construction depends only on the estimated correlations and the selected threshold; sector metadata are attached afterwards as an external classification against which the resulting topology can be interpreted. This distinction is important because a GICS label reflects a firm’s classified economic activity, whereas a network edge represents estimated return co-movement over the sample period. Sector information is therefore treated as an external descriptive reference rather than as a target that the correlation network is expected to reproduce.

To examine sector alignment at node level, let k_i denote the degree of a non-isolated stock i , and let m_i denote the number of its neighbours belonging to the same GICS sector. Its same-sector neighbour share is defined as

$$s_i = \frac{m_i}{k_i}, \quad k_i > 0.$$

Isolated nodes are excluded from this measure because a neighbour share is undefined when $k_i = 0$. Since the GICS sectors differ substantially in size, s_i is also evaluated relative to a sector-size benchmark. Let $g(i)$ denote the sector of stock i , let $n_{g(i)}$ be the total number of stocks belonging to that sector, and let N denote the total network size. Under random neighbour selection, the corresponding same-sector benchmark is

$$b_i = \frac{n_{g(i)} - 1}{N - 1}.$$

The ratio

$$\frac{s_i}{b_i}$$

therefore provides a simple sector-size-adjusted descriptive measure of whether stock i has more or fewer same-sector neighbours than would be expected from the availability of same-sector stocks alone. It is not interpreted as a formal statistical test.

For each non-isolated node, the extension also records the dominant neighbour sector, defined as the sector or sectors appearing most frequently among its neighbours. Ties are retained explicitly rather than resolved arbitrarily. If the stock’s own GICS sector is among several sectors tied for the maximum neighbour count, the neighbourhood is treated as aligned with the official classification for this descriptive comparison.

A separate rule is used to identify stocks occupying unusually cross-sector-oriented

network positions. A stock is classified in this descriptive group only when three conditions hold simultaneously: its degree is at least five; its same-sector neighbour share does not exceed the corresponding random-neighbour benchmark b_i ; and it has a single, unambiguous dominant neighbour sector that differs from its own GICS sector. Nodes with a tie for the dominant neighbour sector are not assigned to this group. The rule is intentionally transparent and descriptive rather than a formal anomaly-detection model. Accordingly, such stocks are referred to as *cross-sector-oriented* or as occupying *sector-atypical network positions*, rather than as being misclassified.

Finally, the extension exports the underlying graph in reusable form in addition to the summary statistics used in Chapter 4. For the large network at the main threshold, the repository contains a node table, an edge list, threshold-level summary tables, node- and sector-level alignment tables, figures and a data provenance report. Equivalent node and edge exports are also produced for the original 47-stock network at $\tau = 0.5$, allowing both the main experiment and the large-network extension to be reconstructed directly from the repository without requiring the network objects to be regenerated first. The empirical results derived from these outputs, and the comparison between the 47-stock and 474-stock networks, are reported in Chapter 4.

3.17 Reproducibility and Limitations

Every tunable parameter of the study—file paths, the sample period, the stock universe and its sector assignment, the cleaning thresholds, the correlation-threshold grid, the MST distance convention, the centrality quantiles used for labelling, and the random seed—is declared once in `config.py`. A small number of command-line options allow the most frequently varied parameters (sample dates, threshold grid, main threshold, BFS source) to be overridden for a single run without editing the configuration file, and every exported report describes the configuration that actually produced it. This single-source-of-truth design supports reproducible computational research (Peng, 2011) and allows any number quoted in Chapter 4 to be regenerated by re-running the pipeline rather than taken on trust.

Provenance is enforced the same way: a data-source report is written at the end of every run, stating explicitly whether the results come from real Yahoo Finance data or from the offline synthetic self-test. The run behind every number confirmed in this chapter used real Yahoo Finance data, with all 47 requested tickers surviving and 1,759 aligned daily return observations spanning 2019–2025.

Methodological Limitations

Several limitations follow directly from the design choices documented in this chapter and are stated here so that Chapter 4 can be read with them in mind:

- The Pearson correlation coefficient captures linear dependence only; non-linear or tail dependence between returns is not measured.
- Correlations are estimated over one fixed historical window (2019–2025) and are treated as constant within it, even though the true dependence structure is known to vary over time, particularly around market stress.
- The choice of correlation threshold τ is partly arbitrary; results are reported across a grid of values precisely to expose this sensitivity, but no single “correct” threshold exists.
- The threshold-filtered and MST networks are static representations and do not capture how the market’s correlation structure evolves through time.
- The primary empirical universe comprises 47 deliberately selected S&P 500 constituents and is therefore not statistically representative of the entire market. The supplementary 474-stock extension substantially broadens the cross-sectional scope, but it is conditional on near-end-of-sample index membership rather than a time-consistent historical reconstruction of the S&P 500. The two universes should therefore be interpreted as complementary descriptive experiments rather than as fully representative samples of the broader equity market.
- A stock’s network centrality is a statement about its estimated co-movement with the rest of the universe; it has no implied relationship with profitability or investment quality.
- BFS and DFS reveal the topology of the estimated correlation network; they identify structural relationships, not causal ones.
- The illustrative portfolio comparison of Section 3.15 is descriptive and in-sample; it is not a validated or backtested investment strategy.

The large-network extension also differs slightly in aligned calendar length: the original 47-stock analysis uses 1,759 daily return observations, whereas the 474-stock extension uses 1,706 after complete-case alignment. Comparisons between the two networks are therefore descriptive and should not be interpreted as isolating the effect of network size alone.

Chapter 4 takes each of the objects built here—the threshold graphs, the connected components, the BFS and DFS results, the Minimum Spanning Tree, the interpretation

framework and the illustrative portfolios—and interprets their real-data output. This chapter has been concerned only with how those objects were produced, why each design choice was made, and what limitations attach to them.

Chapter 4

Results and Analysis

This chapter reports and interprets the empirical results produced by the pipeline described in Chapter 3. Chapter 3 answered the question of how the analysis was implemented; this chapter answers the question of what it found. Every figure, table and numerical claim below is taken from the real-data run confirmed in the pipeline’s data-source report (DATA SOURCE: REAL YAHOO FINANCE DATA; 47 of 47 requested tickers downloaded; 1,759 aligned daily log-return observations spanning 2019-01-02 to 2025-12-31). No synthetic or self-test output is used in this chapter.

4.1 Overview of the Empirical Analysis

The empirical analysis examines 47 S&P 500 stocks drawn from ten GICS sectors, using daily adjusted closing prices over 2019–2025 (1,759 aligned trading days after the calendar alignment and cleaning described in Sections 3.2–3.4). Daily logarithmic returns are computed for every stock, and the full 1,081-pair Pearson correlation matrix is estimated over this single fixed window. Two complementary filtering procedures are then applied: a threshold-based procedure, evaluated across the grid

$$\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\},$$

with $\tau = 0.5$ designated the main threshold, and the threshold-free Minimum Spanning Tree (MST) of Mantegna (1999). Breadth-First Search (BFS), Depth-First Search (DFS) and connected-component detection are applied to the resulting threshold graphs, while articulation points and bridges are computed separately using NetworkX’s Tarjan-based routines. Their outputs are related throughout to the sector composition of the universe.

The methodology behind every object interpreted here was described in Chapter 3 and is not repeated. The analysis throughout this chapter is exploratory and structural rather than predictive. Graph-theoretic quantities such as degree, betweenness, component membership and MST leaf status describe the position of a stock inside one estimated

correlation network over one historical sample; they are not forecasts and do not measure investment quality. Except where Section 4.10 turns explicitly to an illustrative, in-sample portfolio comparison, nothing in this chapter concerns future returns, and nothing here should be read as identifying attractive stocks or a profitable strategy.

The results should be interpreted within the methodological limitations documented in Section 3.17. Pearson correlation captures linear dependence only, the networks are estimated over fixed historical windows, and the correlation threshold remains a modelling choice whose sensitivity is examined across a grid. The primary 47-stock analysis additionally uses a purposive, sector-balanced universe, while the supplementary large-network extension is conditional on near-end-of-sample S&P 500 membership. These design choices constrain the scope of inference but do not prevent descriptive comparison of the estimated network structures.

The chapter proceeds from the correlation matrix and its sectoral structure to threshold sensitivity, the main threshold network, connected components, BFS, DFS-related structural analysis, the MST, the exploratory interpretation framework and the illustrative portfolio comparison. It then reports the supplementary large-network analysis in Section 4.11, before Section 4.12 synthesises the main findings.

4.2 Correlation Matrix and Sectoral Co-Movement

The 1,081 distinct pairwise Pearson correlations estimated from the cleaned return panel have a mean of +0.3754 (median +0.3548, standard deviation 0.1404) and range from +0.0683 (NVDA–VZ) to +0.8804 (JPM–BAC). None of the 1,081 estimated pairwise correlations is negative. This pattern is consistent with a broad common component in large-capitalisation US equity returns over the sample. The economically relevant variation for the network analysis is therefore the dispersion around this uniformly positive average.

The five strongest pairs in the universe are all same-sector—JPM–BAC (+0.8804), GS–MS (+0.8713), COP–EOG (+0.8658), BAC–WFC (+0.8529) and DUK–SO (+0.8501)—and all ten strongest pairs join stocks belonging to the same GICS sector. This provides an initial indication of sectoral structure before any graph is constructed, consistent with the stock-correlation-network literature (Bonanno et al., 2004). At the opposite end, the weakest relationships include NVDA–VZ, NVDA–AEP, NVDA–JNJ, AMZN–DUK, NVDA–DUK and NVDA–T, all below +0.10, together with SLB–WMT at +0.1179. These values describe the estimation window only and are not forecasts of future dependence.

This gap between within-sector and cross-sector co-movement is systematic. Averaged over the 89 same-sector pairs, mean correlation is +0.5747; averaged over the remaining 992 cross-sector pairs it is +0.3575, a difference of +0.2172. Because this comparison is made directly on the correlation matrix, before any threshold or spanning-tree procedure is applied, it shows that sectoral structure is already present in the statistical input to the

graph algorithms.

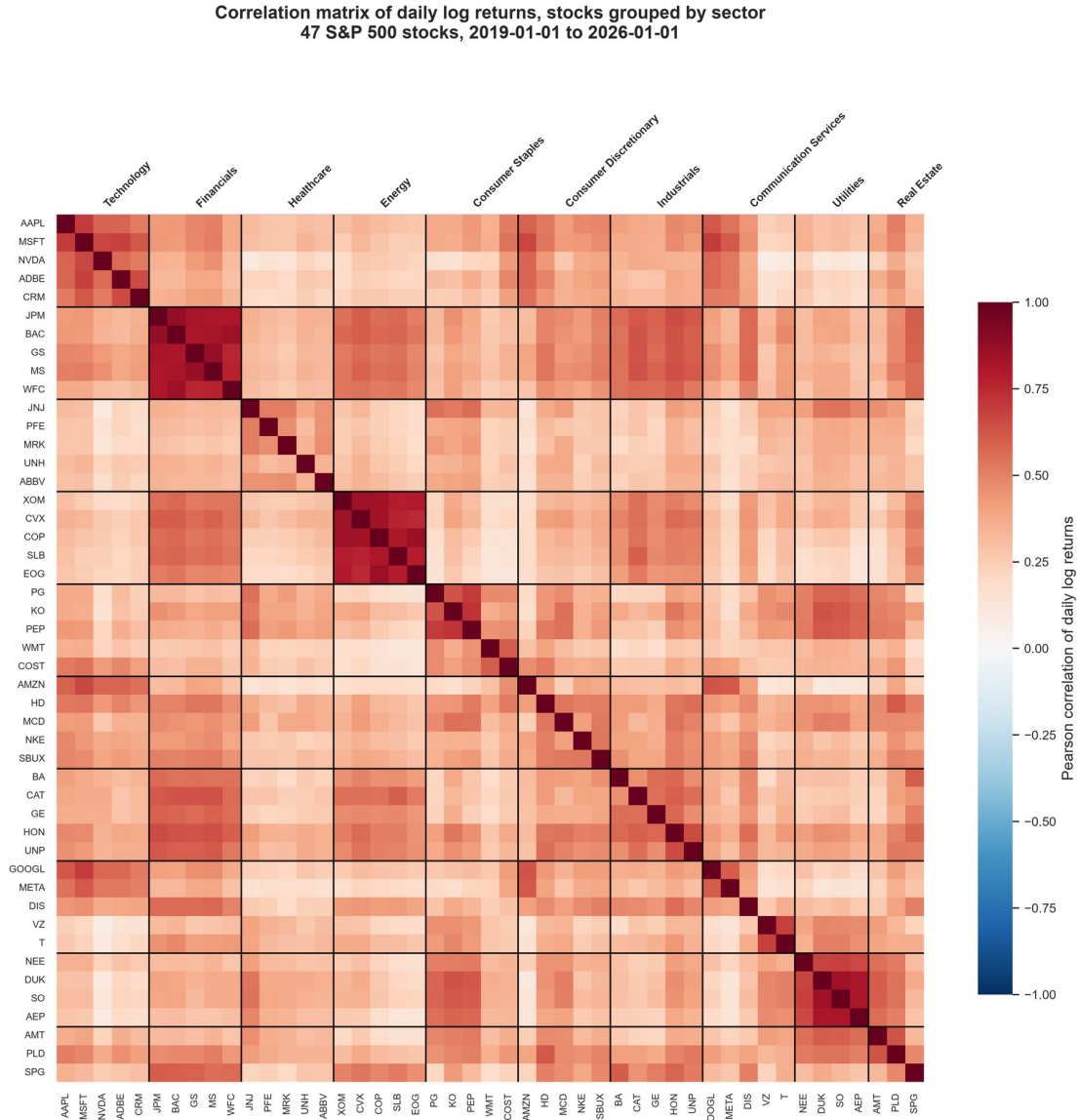


Figure 4.1: Correlation heatmap of the 47-stock universe, with stocks grouped by sector.

Figure 4.1 makes the sectoral pattern visible. The diagonal blocks for Financials and Energy are particularly pronounced relative to much of the off-diagonal area. In numerical terms, within-sector average correlation is highest for Financials (+0.822) and Energy (+0.808), followed by Utilities (+0.748). Healthcare (+0.422) and Communication Services (+0.342) are the least internally cohesive sectors in this sample. Communication Services combines businesses with substantially different economic exposures, which may partly account for its comparatively weak internal co-movement.

Averaged against all 46 other stocks, MS (+0.4762), HON (+0.4730), BAC, JPM and GS have the highest mean correlation with the rest of the universe; WMT (+0.2795), MRK, PFE, VZ and META have the lowest. These lists anticipate some of the threshold-network

results below, which is expected because degree at a fixed threshold is closely related to the number of pairwise correlations that clear that threshold.

Two caveats apply. Pearson correlation measures linear co-movement, not causal influence. Moreover, the matrix is estimated over a single fixed 2019–2025 window containing materially different market environments; a sub-period analysis could produce a different dependence structure (Onnela et al., 2003).

4.3 Threshold Sensitivity and Network Fragmentation

Table 4.1 reports how the threshold-filtered network responds as τ is raised from 0.3 to 0.7.

Table 4.1: Network statistics across correlation thresholds.

τ	Nodes	Edges	Density	Avg. degree	Avg. clustering	Components	Giant component	Isolated
0.3	47	725	0.671	30.85	0.829	1	47 (100.0%)	0
0.4	47	405	0.375	17.23	0.765	2	46 (97.9%)	1
0.5	47	208	0.192	8.85	0.659	3	45 (95.7%)	2
0.6	47	67	0.062	2.85	0.477	15	15 (31.9%)	10
0.7	47	25	0.023	1.06	0.277	35	5 (10.6%)	30

The node set is held fixed at 47 stocks across all threshold levels; increasing τ removes edges rather than vertices, so stocks that lose all surviving connections remain in the graph as isolated nodes.

Raising τ from 0.3 to 0.7 removes 700 of the 725 edges present at the lowest threshold (96.6%), taking density from 67.1% to 2.3%. Average degree falls from 30.85 to 1.06, and average clustering from 0.829 to 0.277.

The most informative feature of Table 4.1 is the abrupt change in the giant component between $\tau = 0.5$ and $\tau = 0.6$. Through $\tau = 0.5$ the largest connected component still contains at least 95.7% of the universe; at $\tau = 0.6$ its share falls to 31.9% and the number of components rises from 3 to 15. A cautious interpretation is that many cross-sector links survive up to the 0.5 cutoff but disappear once the threshold moves into the region of stronger, more sector-specific correlations. By $\tau = 0.7$, 30 of 47 stocks are isolated and the graph has split into 35 components, so the network primarily describes the strongest bilateral and small-cluster relationships rather than overall market connectivity.

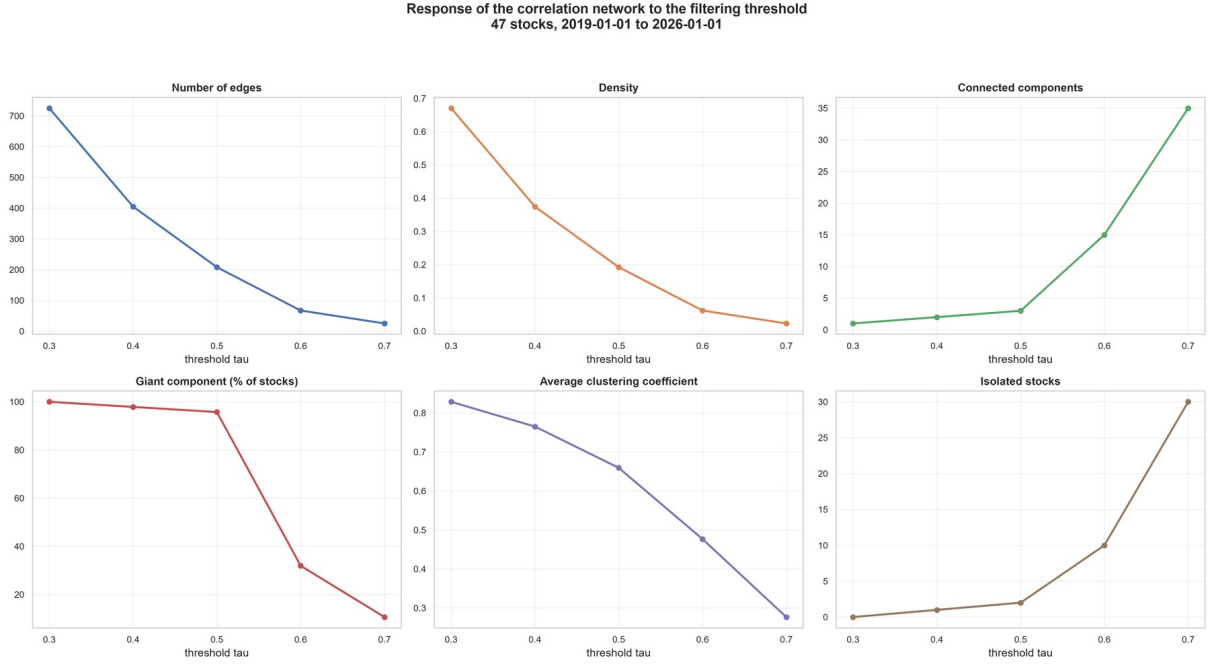


Figure 4.2: Response of the correlation network to the filtering threshold τ .

Figure 4.2 shows the transition visually. The main threshold $\tau = 0.5$ is the last point on the chosen grid before extensive fragmentation: it is strict enough to remove most pairwise links while retaining a giant component containing 45 of 47 stocks. This remains a modelling decision rather than a uniquely correct threshold, which is why the full grid is reported.

4.4 Main Threshold Network at $\tau = 0.5$

At the main threshold, the network built from $|C_{ij}| \geq 0.5$ contains all 47 stocks and 208 edges, with density 0.192 and average degree 8.85. Average clustering is 0.659. For comparison, an Erdős–Rényi graph with connection probability equal to the observed density would have expected clustering approximately equal to $p = 0.192$; the substantially larger observed value is therefore consistent with non-random local cohesion (Newman, 2010). Restricted to the giant component (45 of 47 stocks), the average shortest-path length is 2.518 edges and the diameter is 6 edges.

Intra-sector edges account for 27.4% of the 208 retained edges. Although this may appear low relative to the within-sector correlation advantage in Section 4.2, cross-sector pairs outnumber within-sector pairs by more than 11 to 1 (992 versus 89), so cross-sector pairs still provide most retained edges in absolute terms. Sector assortativity is positive but modest (+0.172), while degree assortativity is also positive (+0.234).

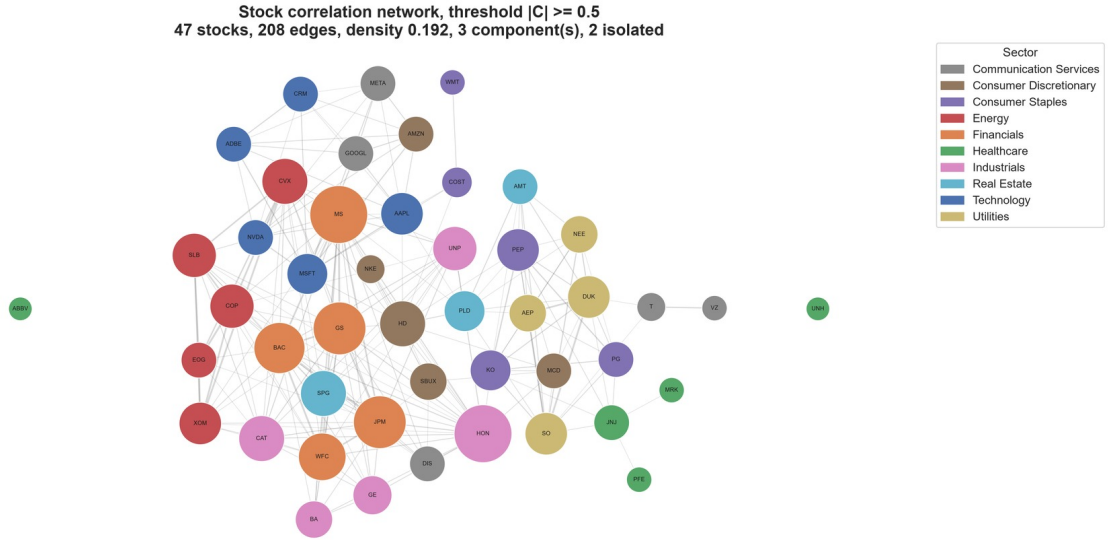


Figure 4.3: Threshold-filtered stock correlation network at $\tau = 0.5$ (node colour = sector; node size proportional to degree).

Figure 4.3 shows a dense core with several lower-degree peripheral chains and two isolated stocks, ABBV and UNH. The force-directed layout is used only as a visual aid; substantive interpretation is based on graph quantities rather than exact geometric position in the drawing.

By composite centrality, the ten highest-ranked stocks are MS, HON, HD, GS, JPM, BAC, PLD, AAPL, WFC and CVX. MS and HON have the highest degree (20 each). Their identical degree masks a small difference in shortest-path brokerage: MS has betweenness 0.176 and closeness 0.540, compared with 0.161 and 0.526 for HON. At the opposite end, ABBV and UNH have degree zero, while VZ, MRK, PFE and WMT each have degree one. These rankings describe structural connectedness within the estimated network, not investment quality.

4.5 Connected Component Analysis

At $\tau = 0.5$ the network has three connected components: a giant component of 45 stocks (95.7%) and two singleton components, UNH and ABBV, both Healthcare stocks.

Seven sectors—Technology, Financials, Energy, Consumer Staples, Consumer Discretionary, Industrials and Communication Services—each contribute exactly five stocks to the giant component; Utilities contributes four, while Real Estate and Healthcare contribute three each. Consequently, no single sector is substantively dominant. The single “dominant sector” label stored by some exported outputs is tie-order dependent (Technology in the component table and Energy in the component figure legend) and is not interpreted as an empirical result.

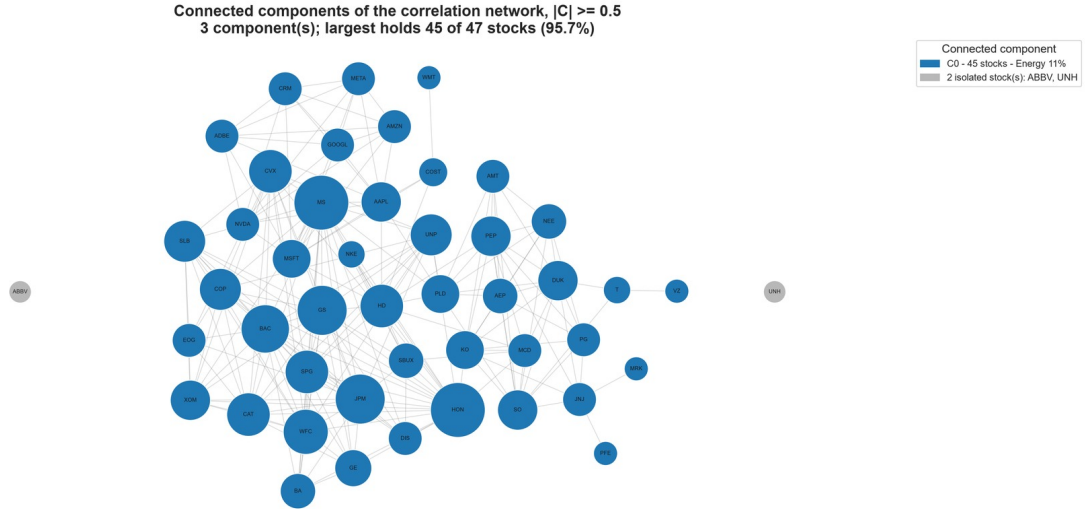


Figure 4.4: Connected components of the correlation network at $\tau = 0.5$. The legend’s sector label for the giant component reflects an arbitrary tie among seven equally represented sectors.

The more accurate description is therefore that the giant component spans all ten sectors in broadly comparable proportions. The market remains structurally integrated at the main threshold even after more than four-fifths of all possible pairwise links have been removed.

UNH and ABBV are the only isolated stocks. Isolation at $\tau = 0.5$ means that neither stock has a correlation of magnitude at least 0.5 with any of the other 46 stocks over this sample. This is a statement about co-movement in the selected universe and historical window, not about either company’s economic importance. Both stocks connect to the market in the threshold-free MST, where all 47 nodes must be spanned.

4.6 Breadth-First Search from AAPL

AAPL is used as the BFS source stock, following the default specified in `config.py`. At $\tau = 0.5$, AAPL has degree 11 and occupies a connected but non-extreme position in the threshold graph.

From AAPL, 44 of the other 46 stocks are reachable; the two unreachable stocks are ABBV and UNH, exactly the two singleton components identified above. AAPL’s eccentricity is 4 edges, and the mean distance to a reachable stock, excluding the source itself, is exactly 2.000 edges. Table 4.2 reports the full layer distribution.

Table 4.2: BFS distance layers from AAPL at $\tau = 0.5$.

Distance	No. of stocks	Example tickers
0 (source)	1	AAPL
1	11	ADBE, MS, MSFT (+8 more)
2	25	BAC, HON, JPM (+22 more)
3	5	EOG, JNJ, KO, PG, T
4	3	MRK, PFE, VZ

AAPL’s eleven direct neighbours combine Technology names with stocks from several other sectors, indicating that its threshold-network neighbourhood is not purely sectoral. More than half of the universe is reached at distance 2. Only EOG, JNJ, KO, PG and T require three steps, while MRK, PFE and VZ require four.

The distance-4 stocks are exactly those connected to the giant component through bridge relationships identified in Section 4.7: MRK and PFE are attached through JNJ, and VZ through T. This provides a consistency check between the independently computed BFS distances and Tarjan-based bridge analysis.

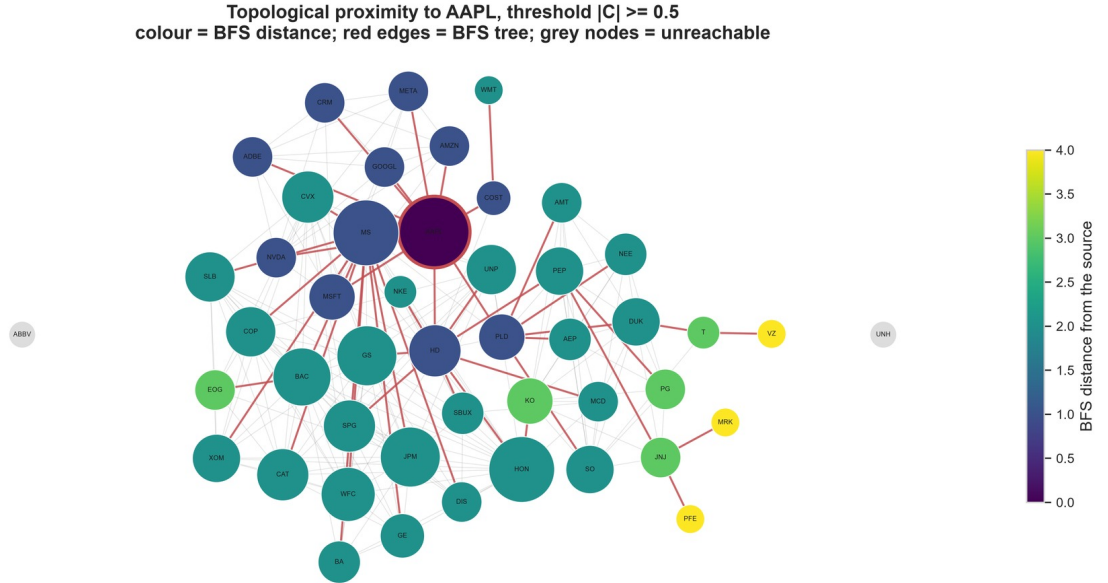


Figure 4.5: Topological proximity to AAPL at $\tau = 0.5$ (node colour = BFS distance; grey nodes = unreachable).

As stated in Section 3.12, the custom BFS implementation was validated node-by-node against NetworkX’s reference routine, with all checks passed. BFS distance measures topological proximity within one estimated network: it is the number of surviving correlation links separating two stocks, not a measure of causal influence or future co-movement.

4.7 Depth-First Search, Articulation Points and Bridges

Restarting the custom single-source DFS from every unvisited node covers the full graph and recovers three connected components, matching NetworkX’s component decomposition. On the same threshold graph, articulation points and bridges are computed separately using NetworkX’s Tarjan-based routines. At $\tau = 0.5$ there are exactly three articulation points and four bridges, summarised in Table 4.3.

Table 4.3: Articulation points and bridges at $\tau = 0.5$.

Type	Element	Effect of removal
Articulation point	JNJ (Healthcare)	Components rise $3 \rightarrow 5$; MRK and PFE become separated from the giant remnant
Articulation point	COST (Consumer Staples)	Components rise $3 \rightarrow 4$; WMT becomes separated from the giant remnant
Articulation point	T (Communication Services)	Components rise $3 \rightarrow 4$; VZ becomes separated from the giant remnant
Bridge	COST–WMT ($C = +0.602$)	Sole link connecting WMT to the giant component
Bridge	JNJ–MRK ($C = +0.509$)	Sole link connecting MRK to the giant component
Bridge	JNJ–PFE ($C = +0.509$)	Sole link connecting PFE to the giant component
Bridge	T–VZ ($C = +0.685$)	Sole link connecting VZ to the giant component

Every bridge joins two stocks from the same sector and isolates exactly one peripheral stock when removed. Removing JNJ has a larger vertex-removal effect than removing COST or T because two bridge-attached stocks, MRK and PFE, lie beyond it. The number of components rises from 3 to 5 and the largest remaining component falls from 45 to 42 stocks.

Structural importance in this sense is not equivalent to high degree. COST has degree 4 at $\tau = 0.5$, far below MS and HON at degree 20, yet it is one of only three articulation points. None of the ten highest composite-centrality stocks is an articulation point. This illustrates the distinction between general connectedness and occupying a unique cut-vertex position.

Across the threshold grid, articulation-point counts are non-monotonic. At $\tau = 0.3$ and 0.4 there are no articulation points or bridges. At $\tau = 0.5$ there are 3 articulation points and 4 bridges; at $\tau = 0.6$, 6 articulation points and 7 bridges; and at $\tau = 0.7$, no articulation points but 2 bridges remain. Once the graph has fragmented mainly into isolated nodes and small components, cut vertices become rare even though individual bridge edges can still exist.

The exact custom DFS parent tree is not used as an empirical result here. The substantive structural conclusions in this section rely on validated component recovery and on the separate Tarjan-based articulation-point and bridge computations.

4.8 Minimum Spanning Tree and Correlation Backbone

The Minimum Spanning Tree retains exactly 46 of the 1,081 possible pairwise correlations (4.3%), the smallest number of links capable of connecting all 47 stocks without a cycle. It is extracted from the Mantegna distance

$$d_{ij} = \sqrt{2(1 - C_{ij})}$$

without choosing a correlation threshold.

The total tree length is 37.5194. Its normalised tree length,

$$L_{\text{norm}} = \frac{1}{N-1} \sum_{(i,j) \in E_{\text{MST}}} d_{ij},$$

is 0.8156, which is simply the mean Mantegna distance across the 46 tree edges. The mean correlation of an MST edge is +0.6550, substantially above the +0.3754 average of the full correlation matrix. The weakest link retained is UNH-PEP at +0.3905, below the $\tau = 0.5$ threshold, which explains why UNH can be isolated in the threshold graph while still connecting to the threshold-free MST. The tree diameter is 14 links and its average path length is 6.416, both larger than in the $\tau = 0.5$ giant component because a tree contains no redundant paths.

The highest-degree MST node is MSFT (Technology), with degree 8. Its direct neighbours span Technology (AAPL, ADBE, NVDA), Consumer Discretionary (AMZN, HD), Consumer Staples (COST), and Communication Services (GOOGL, META). MSFT therefore acts as a cross-sector hub in the sparse backbone rather than a purely intra-sector node.

Table 4.4: Selected MST central nodes and example leaves.

Ticker	Sector	MST degree	Leaf?	Note
MSFT	Technology	8	No	Highest-degree hub
JPM	Financials	5	No	Betweenness 0.496
DUK	Utilities	4	No	Betweenness 0.523
PEP	Consumer Staples	4	No	Betweenness 0.241
MS	Financials	4	No	Betweenness 0.128
HD	Consumer Discretionary	3	No	Highest MST betweenness (0.660)
UNH	Healthcare	1	Yes	Attached to PEP, $C = +0.391$
ABBV	Healthcare	1	Yes	Attached to MRK, $C = +0.467$
WMT	Consumer Staples	1	Yes	Attached to COST, $C = +0.602$
CAT	Industrials	1	Yes	Attached to MS, $C = +0.639$

Degree and betweenness do not rank nodes identically. HD has MST degree 3, below MSFT, JPM, DUK, PEP and MS, yet its betweenness is 0.660, the highest in the tree. Because every pair of nodes in a tree has a unique path, HD lies on many shortest paths connecting large branches despite having relatively few direct neighbours.

Twenty-four of the 47 stocks (51.1%) are leaves of the MST, meaning that they have exactly one incident edge in the retained tree. This should not be interpreted as proving that the retained edge is necessarily the stock’s single strongest pairwise correlation in the full correlation matrix: the MST is a globally constrained optimisation and may discard strong edges that would create a cycle.

The tree follows sector lines far more often than expected from the composition of the sample. Twenty-six of its 46 edges (56.5%) are same-sector links. A simple random-pair benchmark, given the sector sizes in the 47-stock universe, is

$$p_{\text{same}} = \frac{\sum_s \binom{n_s}{2}}{\binom{47}{2}} = \frac{89}{1081} = 0.0823.$$

The observed same-sector share is therefore about 6.87 times this benchmark. Since sector labels do not enter the MST construction, this provides strong descriptive evidence that the correlation backbone reflects sectoral structure already present in the return correlations.

Minimum Spanning Tree of the stock correlation network
Mantegna distance $d = \sqrt{2(1-C)}$; 46 links retained; hub = MSFT; 57% of links join two stocks of the same sector
node size = MST degree; grey links = same sector, red links = cross-sector

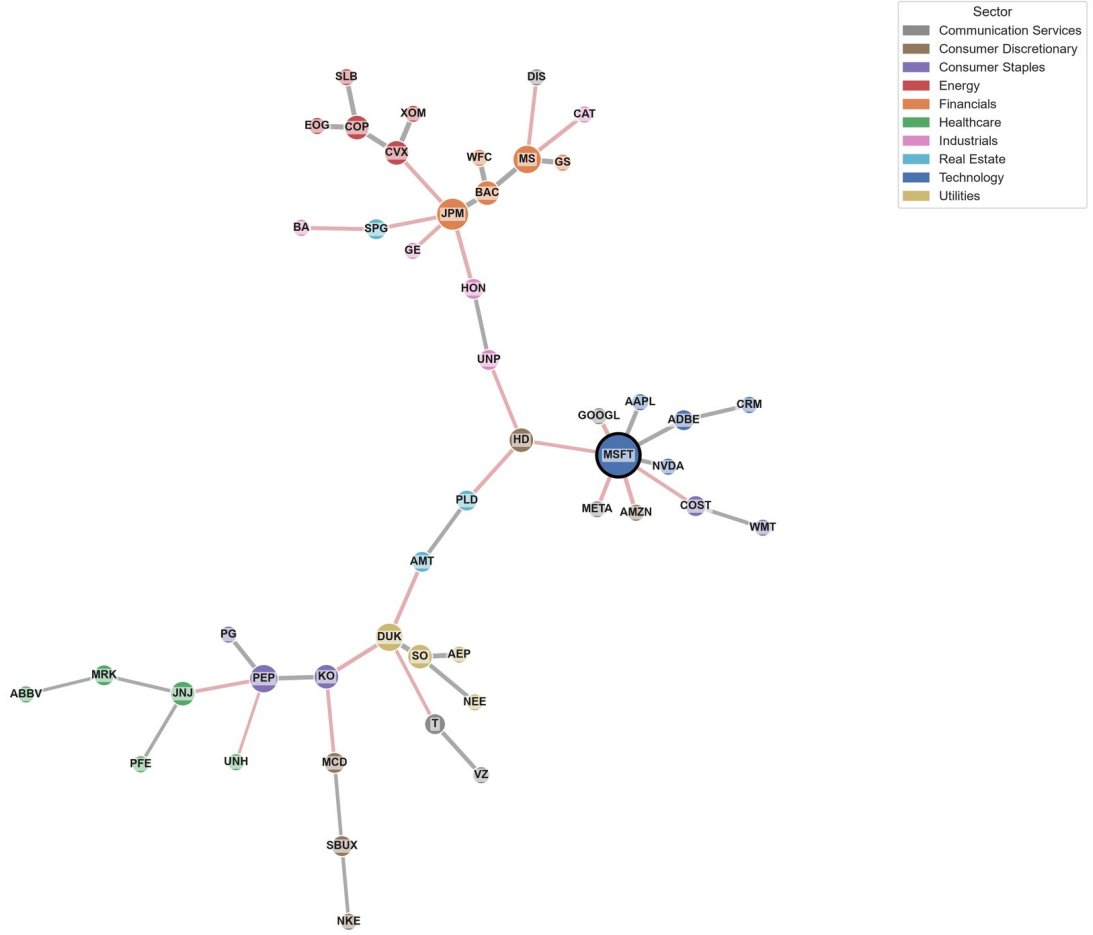


Figure 4.6: Minimum Spanning Tree of the stock correlation network (node size proportional to MST degree; grey links join same-sector stocks and red links join different sectors).

For the sector-level MST summary, define

$$q_s = \frac{I_s}{I_s + E_s},$$

where I_s is the number of internal MST edges within sector s and E_s is the number of MST edges connecting that sector to other sectors. Energy has the highest value ($q_s = 0.80$) and all four possible internal edges required to connect its five stocks. Utilities, Technology and Financials also form internally connected sector branches in the tree. At the other extreme, Communication Services and Real Estate have $q_s = 0.20$, while Industrials has the lowest value (0.17).

4.9 Exploratory Investment Interpretation Framework

Section 3.14 described descriptive, non-exclusive flags attached to every stock: centrality and peripherality based on the 80th and 25th percentiles of composite centrality, bridge status based on articulation-point membership, and MST leaf status. These are combined into a single headline network role. Table 4.5 reports the resulting distribution.

Table 4.5: Exploratory network role classification at $\tau = 0.5$.

Role	N (share)	Example tickers	Note
Hub / central	10 (21%)	AAPL, BAC, CVX, GS, HD, HON, JPM, MS, PLD, WFC	Mean degree 15.1
Intermediate	11 (23%)	AMT, COP, DUK, KO, MCD, MSFT, PEP, SBUX, SO, SPG, UNP	Mean degree 10.2
Peripheral	21 (45%)	ADBE, AMZN, CRM, GOOGL, META, NVDA, PFE, VZ, WMT (+12 more)	Mean degree 6.6
Bridge	3 (6%)	COST, JNJ, T	Same articulation points as Section 4.7
Isolated	2 (4%)	ABBV, UNH	Mean degree 0.0

The three “Bridge” stocks are exactly the three articulation points reported in Section 4.7. This is not independent corroboration: the role classification explicitly defines bridge status using the articulation-point result.

MSFT provides a useful illustration of method dependence. It is the highest-degree hub in the MST, yet in the denser $\tau = 0.5$ threshold graph it has degree 10 and is classified as Intermediate rather than Hub/central. Centrality is therefore not an intrinsic company property; it depends on the network representation and structural measure being used.

Every table and log entry produced by the framework carries the disclaimer that the classification is exploratory, does not constitute investment advice and does not predict future returns.

4.10 Illustrative Portfolio Comparison

This section reports the illustrative, in-sample portfolio exercise described in Section 3.15. It is a secondary extension of the thesis rather than its main contribution. No out-of-sample

test is performed, and transaction costs, taxes and liquidity constraints are not modelled.

Three equally weighted, daily-rebalanced portfolios are compared over the same 1,759 observations:

- **Sector-diversified:** MSFT, BAC, JNJ, COP, PEP, HD, HON, DIS, SO and PLD. One stock is selected from each sector using the deterministic, non-topological rule described in Section 3.15.
- **Network-diversified:** UNH, META, ABBV, WMT, EOG, VZ, PLD, DIS, GE and JNJ. The selection begins from the highest peripherality score (low composite centrality, low degree and MST-leaf status), prioritises network components not yet represented, then minimises the maximum absolute correlation with stocks already selected, with peripherality used as the final tie-breaker.
- **Full universe:** all 47 stocks, equally weighted, as an unconstrained benchmark.

Table 4.6: Illustrative portfolio comparison (equally weighted, in-sample, no transaction costs).

Portfolio	Avg. pairwise corr.	Ann. volatility	Ann. return	Sharpe ($r_f = 0$)	Max drawdown
Sector-diversified	0.438	19.70%	15.42%	0.783	−37.28%
Network-diversified	0.286	18.27%	17.90%	0.980	−32.81%
Full universe (47)	0.375	19.11%	17.92%	0.938	−36.33%

The most relevant result for the purpose of this thesis is average pairwise correlation. The network-diversified portfolio has an average internal correlation of 0.286, below both the sector-diversified portfolio (0.438) and the full-universe benchmark (0.375). It contains stocks from six sectors and spans all three $\tau = 0.5$ connected components, including both ABBV and UNH. The sector-diversified portfolio, by contrast, consists entirely of stocks belonging to the giant component. Within this historical sample, network peripherality therefore produces a basket with lower internal correlation than the sector-representative benchmark, consistent with the diversification interpretation discussed by Pozzi et al. (2013).

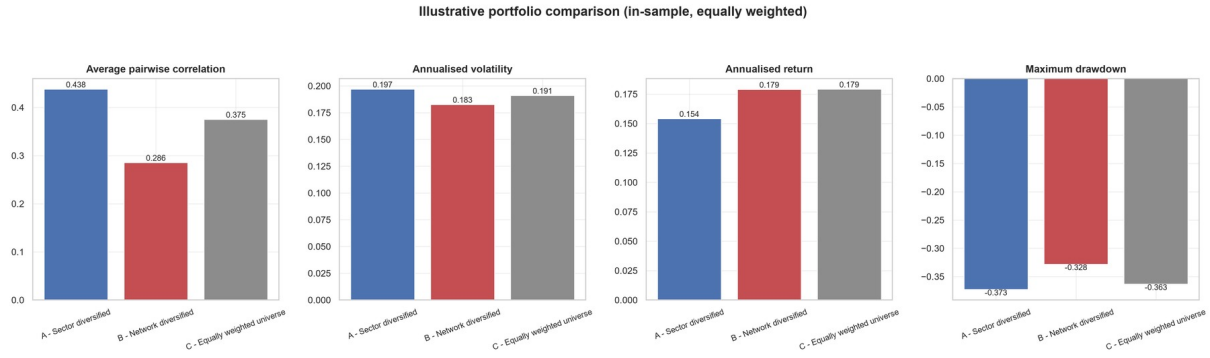


Figure 4.7: Illustrative portfolio comparison: average pairwise correlation, annualised volatility, annualised return and maximum drawdown.

The return, Sharpe-ratio and drawdown statistics are intentionally given less interpretive weight than the correlation result. Over this sample, the network-diversified portfolio also has a higher annualised arithmetic return (17.90% versus 15.42%), a higher Sharpe ratio (0.980 versus 0.783), and a shallower maximum drawdown (-32.81% versus -37.28%) than the sector-diversified portfolio. These differences should not be read as evidence that lower correlation causes higher risk-adjusted return. Classical diversification theory concerns the effect of covariance on portfolio variance, not a guarantee of superior future performance (Markowitz, 1952). The observed ranking belongs to this single in-sample period.

4.11 Large-Network Extension: Scalability and Sector Structure

This section reports the results of the supplementary large-network extension described in Section 3.16. The purpose of the experiment is to examine whether the main structural and sectoral patterns observed in the deliberately compact 47-stock network remain visible when the same correlation-network framework is applied at a substantially larger scale. The extension does not replace the original experiment, which remains the thesis’s primary controlled empirical analysis; instead, it provides an additional scalability and robustness check using a separately defined stock universe.

Of the 503 end-of-sample S&P 500 constituents initially requested, historical prices for 495 were successfully downloaded from Yahoo Finance. After applying the same missing-data criterion used in the main analysis, 474 stocks across 11 GICS sectors were retained. The cleaned panel contains 1,707 aligned price dates and 1,706 daily logarithmic-return observations, compared with 1,759 return observations in the 47-stock experiment. The shorter common calendar results from complete-case alignment across a much larger number of securities and is therefore a methodological comparability point rather than

a change in the underlying correlation methodology. The resulting 474×474 Pearson correlation matrix contains 112,101 distinct stock pairs.

Table 4.7 reports the response of the large network to the same threshold grid used in the main experiment, $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$.

Table 4.7: Threshold sensitivity of the large-network extension (474 stocks).

τ	Nodes	Edges	Density	Avg. degree	Avg. clustering	Components	Giant component	Isolated	Same-sector edge share	Sector assortativity
0.3	474	77,187	0.6885	325.68	0.8618	3	472 (99.6%)	2	13.6%	+0.0182
0.4	474	44,626	0.3981	188.30	0.7732	9	466 (98.3%)	8	17.8%	+0.0487
0.5	474	17,106	0.1526	72.18	0.6514	44	423 (89.2%)	41	28.3%	+0.1371
0.6	474	4,041	0.0360	17.05	0.4858	118	338 (71.3%)	103	57.0%	+0.4572
0.7	474	947	0.0084	4.00	0.2964	254	97 (20.5%)	212	95.1%	+0.9350

As in the 47-stock analysis, increasing τ progressively removes weaker correlation links and fragments the network. Between $\tau = 0.3$ and $\tau = 0.7$, the number of retained edges falls from 77,187 to 947, density declines from 0.6885 to 0.0084, and average degree falls from 325.68 to 4.00. Over the same threshold range, the giant component contracts from 472 stocks (99.6% of the universe) to 97 stocks (20.5%), while the number of isolated stocks rises from 2 to 212. This is consistent, at a substantially larger scale, with the qualitative threshold-fragmentation pattern documented for the 47-stock network in Section 4.3.

The more informative result, however, concerns the sectoral composition of the links that survive as the threshold becomes stricter. The proportion of retained edges connecting stocks belonging to the same GICS sector rises from 13.6% at $\tau = 0.3$ to 95.1% at $\tau = 0.7$. Sector assortativity follows the same pattern, increasing from +0.0182 to +0.9350, as shown in Figure 4.8.

This sectoral pattern is not imposed by the graph-construction procedure. GICS labels are introduced only after the threshold networks have been constructed; the existence of an edge depends exclusively on whether $|C_{ij}| \geq \tau$. The results therefore indicate that, as progressively weaker correlations are removed, the strongest surviving co-movement relationships become increasingly concentrated within recognised sector boundaries. This should not be interpreted as evidence that sector membership causes return co-movement, nor as evidence that the network perfectly reproduces the official GICS classification. At $\tau = 0.7$, for example, the giant component contains only 20.5% of the full universe and some surviving links still connect different sectors. The more limited interpretation is that sectoral organisation becomes increasingly visible among the strongest estimated correlations.

At the main threshold $\tau = 0.5$, the large network contains 474 nodes and 17,106 edges, corresponding to a density of 0.1526 and an average degree of 72.18. Average clustering is 0.6514. The graph contains 44 connected components; the giant component includes 423 stocks, or 89.2% of the universe, while 41 stocks are isolated. Sector assortativity is positive at +0.1371.

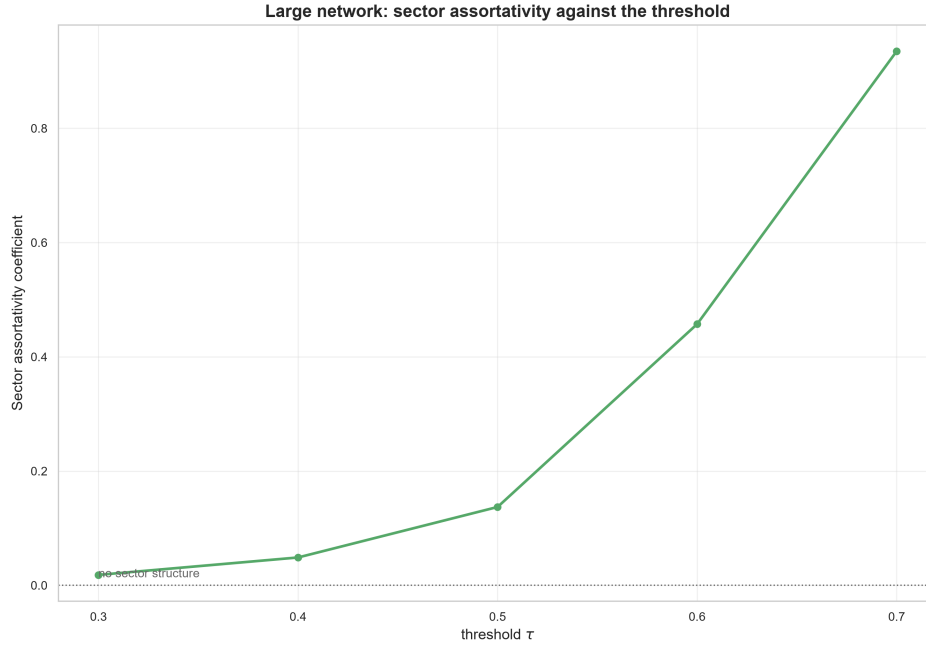


Figure 4.8: Sector assortativity of the 474-stock network across correlation thresholds. Higher thresholds retain progressively stronger correlations and are associated with increasingly sector-concentrated network structure.

The sector-alignment analysis provides an additional perspective on this topology. Among the 433 non-isolated stocks, 28.3% of retained edges connect stocks in the same GICS sector. The corresponding random-pair benchmark implied by the sector composition of the 474-stock universe is 10.8%, so same-sector links occur about 2.63 times as frequently as this composition-based benchmark. The mean same-sector neighbour share across connected stocks is 0.4087 and the median is 0.3133. These values suggest that sector membership remains an important, but far from exclusive, feature of local network structure.

At the level of dominant neighbour sectors, 264 of the 433 non-isolated stocks (61.0%) have their own GICS sector among the sector or sectors most frequently represented in their neighbourhood. Seventeen stocks exhibit a tie for the dominant neighbour sector; in 13 of these cases, the stock's own sector is among the tied leaders. These cases are already included in the 264 aligned observations, since ties containing the official sector are treated as aligned by construction. Sector metadata therefore provide a useful external descriptive reference, but the network does not reproduce the external classification mechanically or uniformly across all stocks.

A direct comparison with the original network at the same main threshold is reported in Table 4.8.

The increase in scale is substantial: the extension contains more than ten times as many stocks as the original experiment. Despite this difference, several descriptive quantities remain close. Average clustering changes only from 0.659 to 0.6514, while the same-sector

Table 4.8: Comparison between the original 47-stock network and the large-network extension at $\tau = 0.5$.

	Original network 47 stocks	Large extension 474 stocks
Nodes	47	474
Edges	208	17,106
Density	0.192	0.1526
Average degree	8.85	72.18
Average clustering	0.659	0.6514
Giant component	45 (95.7%)	423 (89.2%)
Isolated nodes	2	41
Same-sector edge share	27.4%	28.3%
Sector assortativity	+0.172	+0.1371

edge share changes from 27.4% to 28.3%. Sector assortativity also remains positive, although it decreases from +0.172 to +0.1371. At the same time, the large network has lower density and a smaller giant-component share, showing that connectivity is more fragmented when the universe is expanded.

The similarity in clustering and same-sector edge proportions across the two network sizes provides supportive evidence that these particular patterns are not solely artefacts of the deliberately small, sector-balanced 47-stock universe. This comparison is descriptive rather than a formal statistical robustness test, and it does not imply that every property of the original network is invariant to sample composition or scale.

The larger universe also makes it possible to identify stocks whose correlation neighbourhood differs substantially from their official sector classification. Applying the descriptive cross-sector-oriented rule defined in Section 3.16 identifies nine such stocks at $\tau = 0.5$. Table 4.9 reports six illustrative cases.

Table 4.9: Selected cross-sector-oriented nodes in the large network at $\tau = 0.5$.

Ticker	Company	Official sector	Degree	Same-sector share	Dominant neighbour sector
AMZN	Amazon	Consumer Discretionary	25	0.000	Information Technology
SYN	Sysco	Consumer Staples	120	0.008	Financials
GRMN	Garmin	Consumer Discretionary	55	0.018	Financials
UHS	Universal Health Services	Health Care	80	0.063	Financials
TDY	Teledyne Technologies	Information Technology	131	0.076	Financials
SYK	Stryker Corporation	Health Care	138	0.109	Financials

Amazon provides the clearest illustration of this descriptive distinction. Although AMZN is classified within Consumer Discretionary, none of its 25 neighbours in the $\tau = 0.5$ network belongs to that sector, while 68% belong to Information Technology. Its estimated correlation neighbourhood is therefore strongly oriented towards technology stocks despite its formal sector classification. This does not imply that the GICS classification is incorrect: the two objects describe different phenomena, since GICS reflects classified business activity

whereas the correlation network reflects historical return co-movement.

Sysco (SY), Garmin (GRMN), Universal Health Services (UHS), Teledyne Technologies (TDY) and Stryker (SYK) display the same broad cross-sector pattern to varying degrees, with same-sector neighbour shares between 0.8% and 10.9%. For these five cases, Financials is the dominant neighbour sector. These stocks are therefore better described as occupying sector-atypical network positions than as being anomalous or misclassified. The classification rule is deliberately descriptive and does not provide statistical evidence that the observed cross-sector orientation is persistent outside the sample period.

Taken together, the large-network results are broadly consistent with the main structural findings reported for the 47-stock network earlier in this chapter. Threshold sensitivity and the increasing sector concentration of surviving strong correlations remain visible when the analysis is scaled from 47 to 474 stocks. At the same time, the broader universe reveals heterogeneity that is less apparent in the deliberately compact original sample, including a small set of stocks with strongly cross-sector-oriented neighbourhoods.

The extension does not, on its own, validate claims about the broader equity market beyond the universes actually analysed here, and none of its statistics carry causal, predictive or investment significance. Within these limits, the persistence of similar clustering and sector-link patterns at a substantially larger scale provides evidence that the principal structural findings are not solely an artefact of the size or sector-balanced composition of the original 47-stock universe.

4.12 Synthesis of Empirical Findings

The empirical analysis can be summarised in nine main findings:

1. The correlation matrix contains a pronounced sectoral layer. Mean within-sector correlation is +0.575 versus +0.358 across sectors, a difference of 0.217, and all ten strongest pairs are same-sector.
2. Threshold filtering produces a strongly non-linear fragmentation pattern. In the original 47-stock network, the giant component contains more than 95% of the universe through $\tau = 0.5$ but falls below one third at $\tau = 0.6$.
3. At $\tau = 0.5$, 45 of the 47 stocks in the original network remain in one sector-mixed giant component, while ABBV and UNH are isolated. No single sector dominates the giant component.
4. BFS from AAPL reaches 44 of the other 46 stocks within four hops. The most distant reachable stocks are those lying behind the bridge relationships identified by the separate articulation-point and bridge analysis.

5. The custom DFS recovers the same connected components as NetworkX, while the separate Tarjan-based analysis identifies three articulation points (JNJ, COST and T) and four bridges at the main threshold. Articulation-point counts are non-monotonic across the threshold grid.
6. The threshold-free MST contains 46 edges, 56.5% of which are same-sector links, compared with an 8.2% random-pair benchmark. MSFT is the highest-degree MST hub, whereas HD has the highest MST betweenness.
7. The exploratory role framework classifies 10 stocks as Hub/central, 11 as Intermediate, 21 as Peripheral, 3 as Bridge and 2 as Isolated. MSFT's different role in the MST and threshold graph illustrates that centrality is representation-dependent.
8. The illustrative network-diversified portfolio has a lower average pairwise correlation (0.286) than both the sector-diversified benchmark (0.438) and the full 47-stock universe (0.375). This comparison is descriptive and entirely in-sample.
9. The supplementary large-network extension shows that several of the principal structural patterns remain visible when the universe is expanded from 47 to 474 stocks. At $\tau = 0.5$, the large network contains 17,106 edges, has average clustering of 0.6514 compared with 0.659 in the original network, and has a same-sector edge share of 28.3% compared with 27.4% in the original sample. Sector assortativity remains positive at +0.1371, and increasingly strict thresholds leave progressively more sector-concentrated surviving links. The comparison is descriptive rather than a formal statistical robustness test.

Taken together, these findings support the thesis's main empirical argument: graph-based analysis—combining threshold networks, BFS and DFS traversal, connected components, articulation-point and bridge analysis, and the MST—makes interpretable structural patterns in stock-return correlations explicit. In the original controlled sample, the analysis reveals sectoral clustering, a highly connected core, peripheral stocks and specific cut-vertex relationships. The supplementary 474-stock extension provides additional evidence that important features of this structure, particularly clustering and sector-related link formation, are not solely consequences of the deliberately small and sector-balanced original universe.

Sector metadata are not used to determine the edges of either the original or large threshold networks and are introduced only afterwards as an external classification for interpretation. The resulting sectoral alignment should therefore be read as a descriptive feature of the estimated correlation topology rather than as evidence that sector membership causes co-movement or that the network must reproduce the official GICS classification.

All results remain subject to the limitations stated in Section 3.17. Pearson correlation captures linear dependence only, the threshold is a modelling choice examined across a grid, and both analyses estimate static networks over fixed historical windows. The 47-stock experiment remains a purposive controlled sample, while the 474-stock extension is conditional on near-end-of-sample S&P 500 membership rather than a time-varying historical reconstruction of the index. The portfolio exercise is also in-sample and descriptive rather than predictive. Chapter 5 develops these limitations into directions for future work, including rolling-window analysis, time-consistent index membership, alternative dependence measures, international equity universes and genuine out-of-sample portfolio evaluation.

Chapter 5

Conclusion and Future Work

5.1 Summary of the Thesis and Main Findings

This thesis investigated whether graph traversal algorithms and graph-based filtering methods can reveal interpretable structural patterns in stock correlation networks. Stocks were represented as nodes and pairwise return correlations as relationships from which threshold-filtered graphs and a Minimum Spanning Tree (MST) were constructed. Breadth-First Search (BFS) was used to study topological proximity, Depth-First Search (DFS) to recover traversal coverage and connected components, and NetworkX’s Tarjan-based routines to identify articulation points and bridges. The MST provided a complementary, threshold-free representation of the strongest positive co-movement backbone.

The objective was not to predict future returns or to develop a trading strategy. Rather, the thesis examined whether classical graph algorithms can make structural features embedded in equity correlation matrices easier to identify and interpret, and what those features may contribute to exploratory financial-network and diversification analysis.

The primary empirical analysis used a purposive, sector-balanced sample of 47 S&P 500 stocks drawn from ten GICS sectors and daily adjusted closing prices over 2019–2025. Daily logarithmic returns were computed and a Pearson correlation matrix was estimated over a single fixed window. Threshold-filtered graphs were evaluated over

$$\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\},$$

with $\tau = 0.5$ used for the detailed single-network analysis, while the MST was constructed from the Mantegna distance (Mantegna, 1999).

To examine whether the principal structural findings remained visible at a much larger scale, the thesis also included a supplementary large-network extension. Starting from 503 near-end-of-sample S&P 500 constituents, 474 stocks across 11 GICS sectors satisfied the stated data-quality requirements and were retained in the final extended network. This experiment used the same threshold grid and correlation-network construction while

treating sector classifications only as external metadata for interpretation.

The main empirical findings can be summarised as follows:

1. The original correlation matrix displayed a pronounced sectoral structure. Mean within-sector correlation was +0.575, compared with +0.358 across sectors, and the ten strongest pairs were all same-sector.
2. Threshold filtering revealed a strongly non-linear fragmentation pattern. In the 47-stock network, the giant component contained more than 95% of the universe through $\tau = 0.5$, but fell below one third at $\tau = 0.6$.
3. At the main threshold, 45 of the 47 stocks belonged to one sector-mixed giant component, while ABBV and UNH were isolated. No single sector substantively dominated the giant component.
4. BFS from AAPL reached 44 of the other 46 stocks within four hops, with a mean distance of exactly two edges among reachable stocks. This indicates a topologically compact giant component at the chosen threshold.
5. The custom DFS recovered the same connected-component structure as NetworkX, while the separate Tarjan-based analysis identified three articulation points—COST, JNJ and T—and four bridges at $\tau = 0.5$ (Tarjan, 1972).
6. The threshold-free MST retained 46 edges and showed pronounced sectoral organisation: 56.5% of MST edges were same-sector, compared with an 8.2% random-pair benchmark. MSFT was the highest-degree MST hub, while HD had the highest MST betweenness.
7. The exploratory role-classification framework distinguished central, intermediate, peripheral, bridge and isolated stocks. In the illustrative portfolio exercise, the network-diversified portfolio had a lower average pairwise correlation than both the sector-diversified portfolio and the equally weighted full 47-stock universe. This result is descriptive and entirely in-sample.
8. The supplementary 474-stock extension showed that several of the principal structural patterns remained visible at a substantially larger scale. At $\tau = 0.5$, average clustering was 0.6514, close to 0.659 in the original network, while the same-sector edge share was 28.3% compared with 27.4% in the 47-stock network. Sector assortativity remained positive at +0.1371, and progressively stricter thresholds produced increasingly sector-concentrated surviving links. The extension also identified a small set of stocks occupying cross-sector-oriented network positions. These comparisons are descriptive rather than formal statistical robustness tests.

Taken together, the results are consistent with the broader literature showing that stock-correlation networks contain economically recognisable sectoral and hierarchical structure (Mantegna, 1999; Onnela et al., 2003). In both the main experiment and the large-network extension, sector classifications are used for interpretation rather than to determine which threshold-network edges exist. The persistence of similar clustering and sector-link proportions in the substantially larger network therefore provides additional evidence that these patterns are not solely consequences of the deliberately compact and sector-balanced design of the original 47-stock sample.

5.2 Contribution and Practical Implications

The contribution of this thesis is not the invention of a new graph algorithm. BFS, DFS, connected-component detection, Tarjan’s articulation-point method and MST construction are long-established techniques. The contribution instead lies in integrating these methods into a clear and reproducible empirical framework for analysing stock correlation networks, and in showing how their outputs can be interpreted in a finance-related setting.

More specifically, the thesis makes six contributions.

First, it places classical graph traversal algorithms at the centre of the empirical analysis rather than treating them only as hidden computational subroutines. BFS and DFS are implemented explicitly, their outputs are documented, and the custom routines are validated against NetworkX reference procedures where appropriate.

Second, it shows how BFS distances, DFS-based component recovery and threshold sensitivity can be used jointly to describe topological proximity, cohesion and fragmentation in an equity correlation network. This makes the algorithmic logic behind commonly reported network properties visible rather than leaving it implicit.

Third, the analysis demonstrates that articulation-point and bridge detection can identify structurally important cut-vertex and cut-edge relationships that degree or general centrality measures do not capture on their own. These results provide a complementary view of network structure by distinguishing general connectedness from occupying a unique connectivity role.

Fourth, the thesis uses the Minimum Spanning Tree as a threshold-free complement to the threshold-filtered graphs. The MST provides a sparse correlation backbone and shows that a substantial share of its retained links follows recognised sector boundaries even though sector labels are not used in constructing the tree.

Fifth, the supplementary large-network extension tests whether selected structural and sectoral patterns remain visible when the analysis is scaled from 47 to 474 stocks. Similar average clustering and same-sector edge proportions persist descriptively at the larger scale, while increasingly strict thresholds produce progressively more sector-concentrated surviving relationships. The extension also uses sector metadata as an external descriptive

reference to identify stocks whose correlation neighbourhood is unusually cross-sector-oriented. This does not constitute a formal statistical robustness test, but it reduces the concern that the main structural findings are purely consequences of the deliberately compact 47-stock universe.

Sixth, the complete analysis is implemented as a reproducible Python pipeline with centralised configuration, explicit data provenance, validation checks and reusable network exports. The original 47-stock network and the supplementary large network are both exported in node-table and edge-list form, allowing the graph objects used in the empirical analysis to be inspected or reused independently of the original execution environment.

The practical implications are deliberately limited. The framework does not produce buy or sell recommendations, does not identify investment quality and does not establish causal relationships between stocks. Instead, it supports structural interpretation. An analyst can examine which stocks occupy central or peripheral positions, how network cohesion changes as correlation requirements become stricter, which nodes or edges occupy unique connectivity roles, whether strong correlations become increasingly sector-concentrated, and which stocks display correlation neighbourhoods that differ from their formal sector classification.

The illustrative portfolio comparison provides one example of how such structural information may be related to diversification analysis, but it remains entirely in-sample and is not a validated investment strategy. In this sense, graph analysis is best treated as a decision-support and exploratory layer rather than as a complete financial model. Network position may complement established financial analysis, but it does not replace valuation, risk modelling, portfolio optimisation or genuine out-of-sample validation.

5.3 Limitations of the Study

The conclusions of the thesis should be interpreted within several limitations that follow directly from the research design.

1. **Pearson correlation.** The analysis captures linear co-movement only. Non-linear dependence, tail dependence and asymmetric relationships between returns are not measured. The estimated correlations are also sensitive to the chosen historical window and to unusual observations.
2. **Static network.** The correlation matrices and the networks derived from them are estimated over fixed historical windows rather than through a rolling or dynamic framework. The analysis therefore does not model how correlations, network topology, centrality or component structure evolve through time (Onnela et al., 2003).
3. **Threshold sensitivity.** The main threshold $\tau = 0.5$ is a modelling choice. The full grid from 0.3 to 0.7 is reported precisely to expose this sensitivity, but no single

threshold is uniquely correct. Structural conclusions such as component membership, articulation status and isolation may change as the filtering requirement changes.

4. **Stock-universe design and index membership.** The primary 47-stock analysis uses a purposive, sector-balanced sample designed for controlled comparison rather than a statistically representative sample of the full equity market. The supplementary 474-stock extension reduces this particular concern by showing that several structural and sectoral patterns remain visible in a much broader universe. However, the extension is constructed from a near-end-of-sample S&P 500 membership snapshot and applies that universe retrospectively over 2019–2025. It is therefore conditional on end-of-period membership and should not be interpreted as a survivorship-bias-free reconstruction of the historical S&P 500.
5. **Comparability of the two empirical universes.** The original and extended analyses do not differ only in node count. The 47-stock network contains 1,759 aligned return observations, whereas the 474-stock extension contains 1,706 because complete-case alignment across the larger panel removes additional dates. Universe composition also differs substantially. The similarity of selected network statistics across the two analyses is therefore supportive descriptive evidence rather than a controlled experiment isolating the causal effect of network size.
6. **Sector metadata as an external reference.** GICS classifications are used only after graph construction to interpret sectoral alignment. They provide a recognised external classification, but they are not a statistical ground truth that the correlation network is expected to reproduce. A stock whose strongest correlation neighbourhood differs from its official sector should therefore not be interpreted as being incorrectly classified.
7. **Correlation is not causation.** Edges represent statistical co-movement over the sample period, not causal influence, information transmission or direct financial exposure. The network therefore describes estimated dependence structure rather than economic or financial transmission channels.
8. **Network position is not investment quality.** A central stock is not necessarily a better investment, a peripheral stock is not necessarily attractive, and an articulation point is not necessarily riskier or more profitable. These labels describe structural positions in an estimated network and should not be interpreted as valuation or forecasting signals.
9. **Illustrative portfolio comparison.** The portfolio exercise is evaluated entirely in-sample. It does not include genuine out-of-sample testing and does not model transaction costs, taxes, liquidity constraints, rebalancing frictions or robustness

across market regimes. It is therefore an illustration of diversification-related network information rather than a validated trading strategy.

These limitations define the scope of inference from the empirical results. They do not invalidate the descriptive findings reported in Chapter 4, but they constrain how far those findings can be generalised, interpreted causally or translated into predictive and investment claims. The large-network extension strengthens the empirical scope of the thesis, but it does not remove the need for time-varying analysis, alternative dependence measures or genuinely out-of-sample validation.

5.4 Future Work

Several extensions follow naturally from the limitations identified above.

A first direction is to introduce a temporal dimension. Rolling-window correlation networks would make it possible to study whether component structure, centrality, articulation points, sector alignment and MST branches remain stable across market regimes or change materially during periods of stress. This would address one of the main limitations of the current static 2019–2025 analysis and would allow the network to be interpreted as an evolving object rather than as a single full-sample representation.

A second direction is to reconstruct the stock universe through time using time-consistent historical index membership. The large-network extension in this thesis uses a near-end-of-sample S&P 500 membership snapshot and therefore provides a broad cross-sectional robustness check rather than a survivorship-bias-free historical reconstruction of the index. A future study could use constituent membership valid on each date, or at regular reconstitution intervals, so that stocks enter and leave the analysis when they actually enter and leave the index. This would provide a stronger basis for studying the evolution of sector structure and network topology over long historical periods.

A third direction is to extend the analysis beyond the U.S. large-cap universe. International equity indices, multiple national markets or broader capitalisation segments could be used to examine whether the sectoral and topological patterns identified here persist under different market structures. Such an extension would also allow cross-market comparisons of fragmentation, sector assortativity and the prevalence of cross-sector-oriented network positions.

A fourth direction is to compare alternative dependence measures and filtering procedures. Pearson correlation could be complemented by rank correlations (Millington and Niranjana, 2021), partial correlations, or measures designed to capture non-linear or tail dependence. Similarly, the MST could be compared with filtering methods such as the Planar Maximally Filtered Graph or the Triangulated Maximally Filtered Graph. Correlations could also be adjusted for a common market factor before graph construction

in order to isolate more sector-specific or idiosyncratic co-movement.

A fifth direction is to develop a more formal statistical evaluation of the sector-alignment patterns identified in the large-network extension. The same-sector edge share, sector assortativity and node-level neighbour alignment used in this thesis are descriptive. Future work could compare these quantities with permutation-based null models, bootstrap procedures or other statistical benchmarks to quantify how unlikely the observed sector concentration would be under appropriate randomised alternatives.

A sixth direction is to integrate graph-derived information with established portfolio construction methods. Network centrality, peripherality, component membership, articulation status, MST leaf status or cross-sector orientation could be supplied as additional structural inputs to mean–variance optimisation (Markowitz, 1952), risk-parity methods, Hierarchical Risk Parity, clustering-based portfolio construction or machine-learning models. Such extensions would require a separate predictive research design with portfolio-formation rules fixed in advance, rolling estimation, genuine out-of-sample evaluation, transaction costs and statistical comparison with appropriate benchmarks. Prior work on network peripherality provides one motivation for this direction (Pozzi et al., 2013).

Finally, the analysis could be exposed through an interactive dashboard that allows a user to change thresholds, select source stocks, inspect traversal results and explore the large-network sector structure without interacting directly with Python code. This would not alter the underlying methodology, but it could make the framework more useful for exploratory analysis and teaching.

5.5 Final Conclusion

Within the scope of the selected data and modelling choices, this thesis shows that classical graph algorithms provide a clear and reproducible way to analyse stock correlation networks. BFS makes topological proximity explicit, DFS supports traversal and component recovery, threshold sensitivity reveals how network cohesion changes as weaker links are removed, and Tarjan-based articulation-point and bridge analysis identifies unique structural dependencies. The Minimum Spanning Tree provides a complementary sparse backbone that preserves strong co-movement relationships without requiring a correlation threshold.

The original 47-stock analysis provides the main controlled empirical setting of the thesis, while the supplementary 474-stock extension shows that several of the principal structural patterns remain visible at a substantially larger scale. In particular, average clustering and same-sector edge proportions remain similar at the main threshold, while increasingly strict thresholds in the large network leave progressively more sector-concentrated surviving relationships. This does not constitute a formal statistical proof of robustness, but it provides additional evidence that the principal findings are not solely artefacts of the

deliberately compact and sector-balanced original universe.

The large-network extension also highlights that official sector classifications and return-correlation neighbourhoods need not coincide perfectly. Sector metadata are therefore most useful as an external descriptive reference rather than as a target that the network is expected to reproduce. Stocks occupying cross-sector-oriented positions illustrate how graph-based analysis can reveal structural relationships that are not captured by sector labels alone.

The broader conclusion is not that graph traversal predicts financial markets, nor that correlation networks identify causal relationships or superior investments. Rather, graph-based methods provide an interpretable computational language for describing estimated dependence structure. Used with appropriate caution, they make complex co-movement patterns easier to inspect, compare and communicate, and they provide a useful foundation for future work involving dynamic networks, time-consistent index membership, alternative dependence measures, formal null models and genuinely out-of-sample portfolio analysis.

Appendix A

Supplementary Figures

This appendix reports four supplementary figures that support the empirical analysis presented in Chapter 4. Each figure complements a discussion already developed in the main text, and the note below each figure indicates the corresponding section.

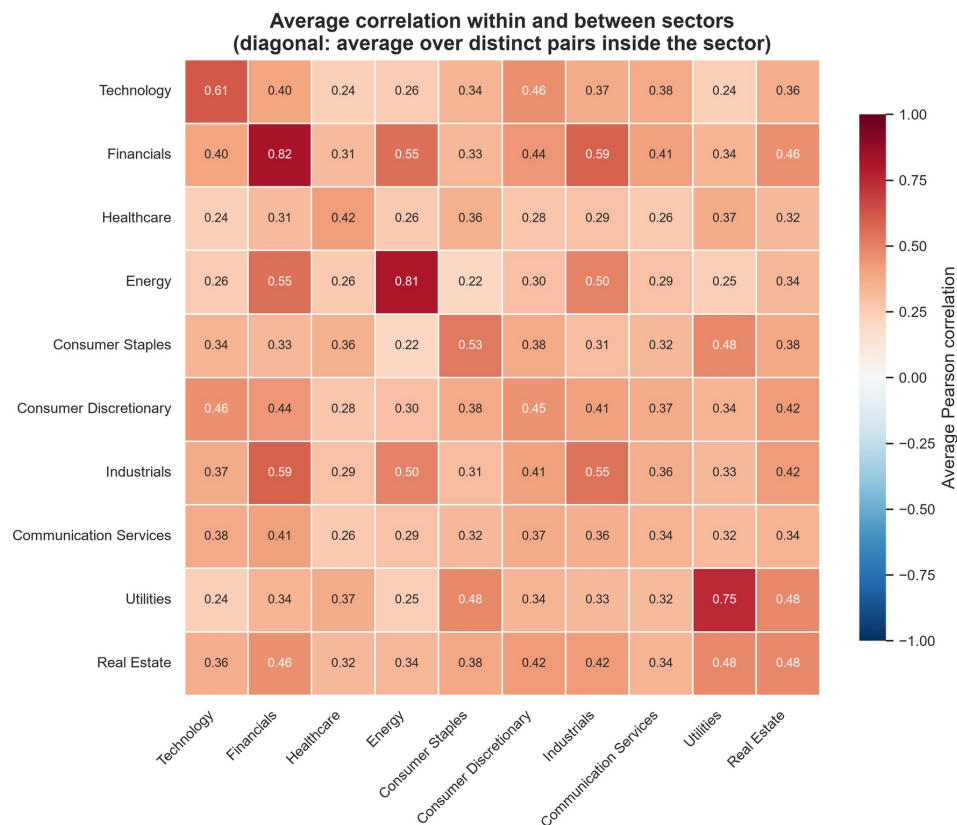


Figure A.1: Sector-average correlation heatmap. The diagonal reports the average correlation between distinct pairs inside each sector, while the off-diagonal cells report average cross-sector correlations.

This figure supplements the correlation-matrix discussion in Chapter 4, Section 4.2.

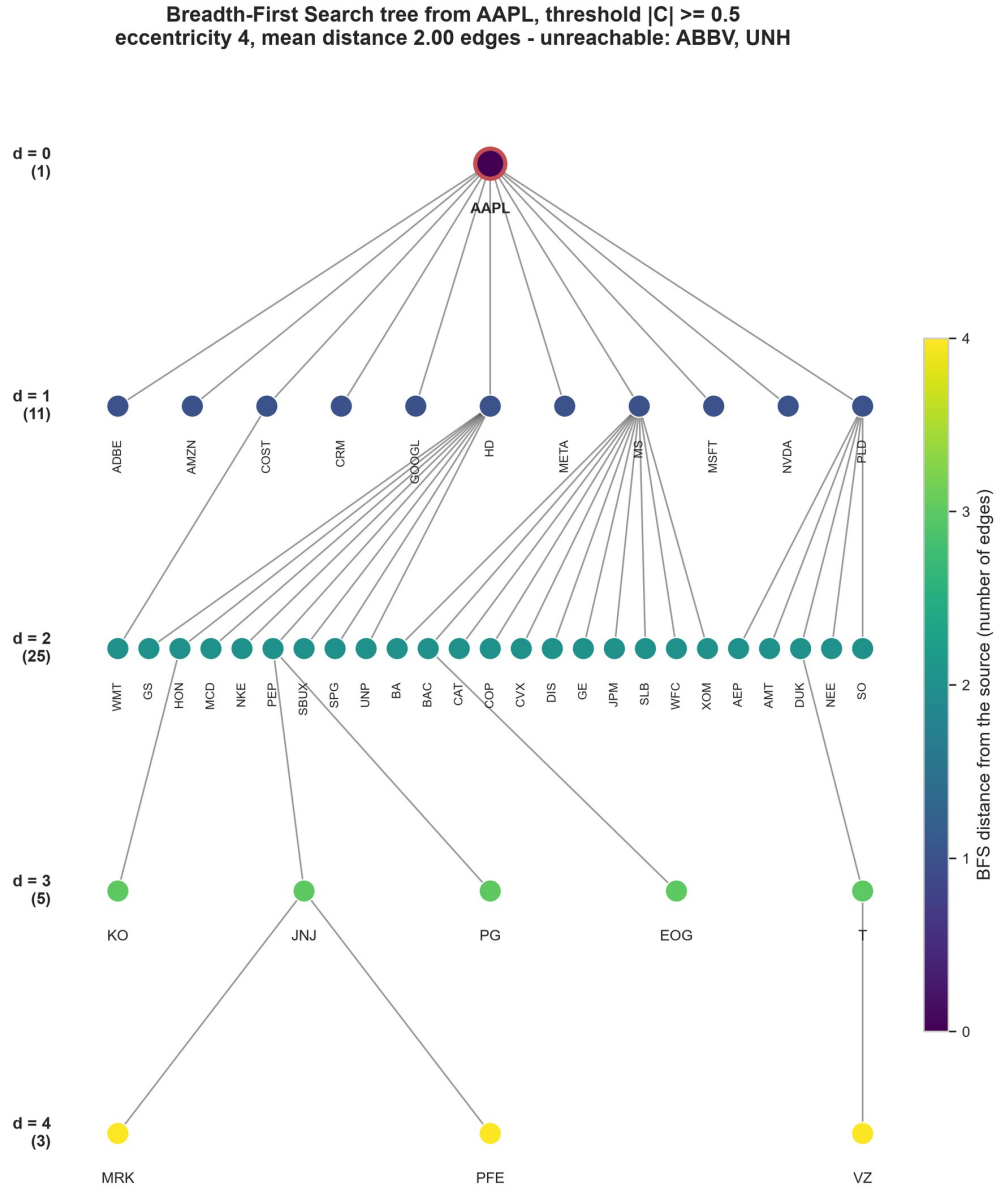


Figure A.2: Breadth-First Search tree from AAPL at $\tau = 0.5$. The figure shows the specific BFS parent of each reachable stock and the layer-by-layer structure of the traversal.

This figure supplements the BFS distance-map discussion in Chapter 4, Section 4.6.

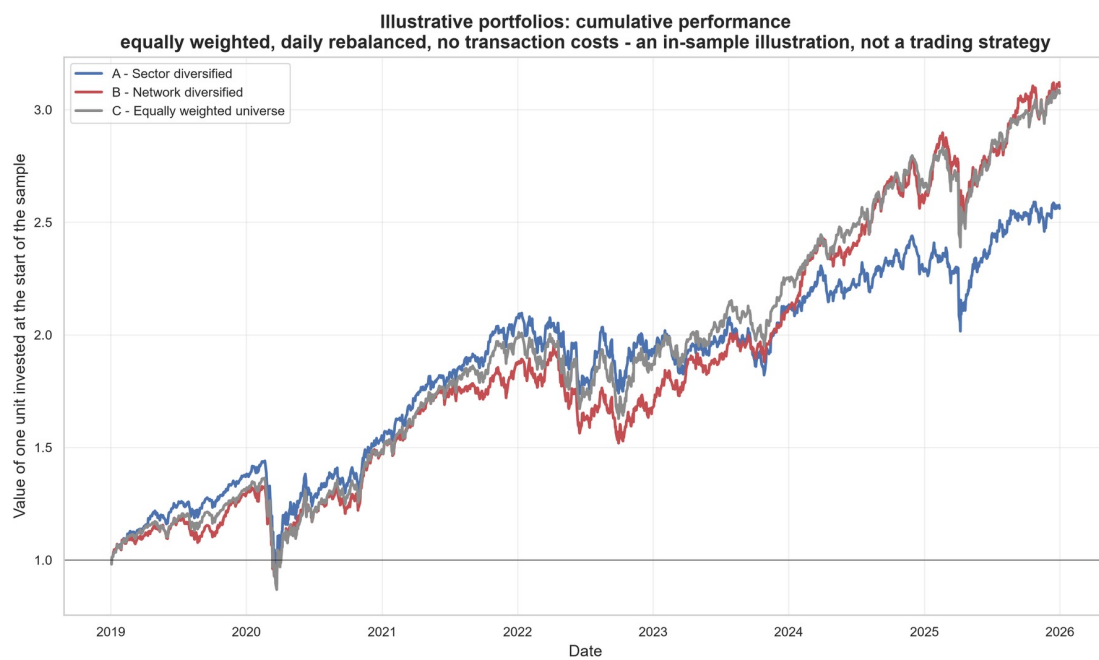


Figure A.3: Illustrative portfolios: cumulative value of one unit invested at the start of the sample. The figure is reported only as supplementary material because the portfolio comparison is descriptive and in-sample.

This figure supplements the illustrative portfolio comparison in Chapter 4, Section 4.10.

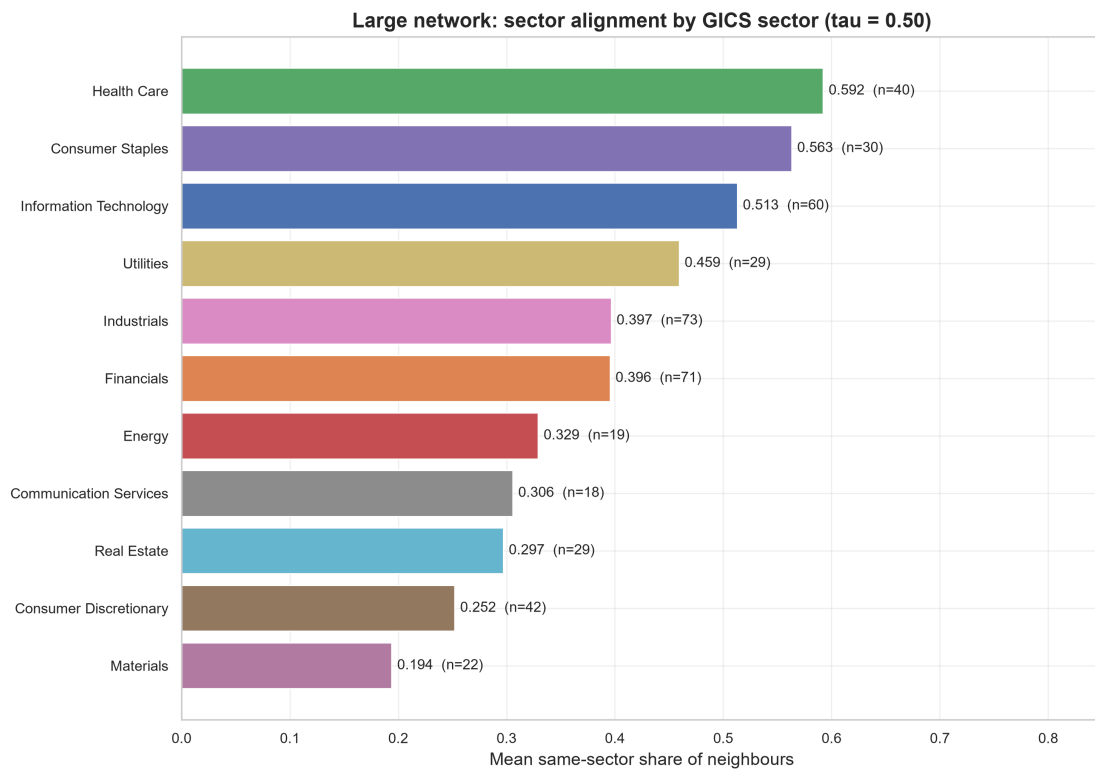


Figure A.4: Mean same-sector neighbour share by GICS sector in the 474-stock network at $\tau = 0.5$. Values are computed over non-isolated stocks and show how strongly local correlation neighbourhoods align, on average, with each stock's external sector classification. *This figure supplements the large-network sector-alignment discussion in Chapter 4, Section 4.11.*

Appendix B

Reproducibility Materials

This appendix summarises the repository structure and the steps required to reproduce the empirical results reported in Chapters 3 and 4. Selected core traversal implementations are reproduced in Section B.5 to document the algorithms most directly related to the computer-science contribution of the thesis. The complete source code is available in the accompanying GitHub repository. A frozen copy of the thesis version is also provided as `thesis_code.zip`.

Source-code repository:

<https://github.com/sofiasabbioni/thesis-network-analysis>

Thesis code version: `v2.0-thesis`

B.1 Repository Structure

Table B.1 summarises the main files and folders of the repository that implements the empirical pipeline.

B.2 Software Environment

The empirical pipeline is implemented in Python. The project file `requirements.txt` specifies minimum compatible versions rather than pinning a single environment: `pandas>=1.5`, `numpy>=1.23`, `yfinance>=0.2.40`, `networkx>=3.0`, `matplotlib>=3.6`, `seaborn>=0.12` and `scipy>=1.9`. SciPy is used for a small number of descriptive statistics, including skewness and excess kurtosis.

The exact interpreter and package versions recorded by the real-data run that generated the Chapter 4 outputs are reported in Table B.2.

Table B.1: Repository structure of the empirical pipeline.

File / folder	Contents
<code>main.py</code>	Main execution script; orchestrates the full pipeline.
<code>large_network_extension.py</code>	Separate execution script for the supplementary 474-stock large-network analysis.
<code>config.py</code>	Analysis parameters: sample period, ticker list (47 stocks, 10 sectors), correlation thresholds and BFS source stock.
<code>requirements.txt</code>	Python dependencies, specified as minimum compatible versions.
<code>README.md</code>	Repository instructions and documentation.
<code>src/</code>	Source modules for data collection, preprocessing, graph construction, traversal algorithms, metrics, MST analysis, portfolio analysis and visualisation.
<code>data/reference/</code>	Frozen S&P 500 membership and GICS metadata snapshots, with provenance information used by the large-network extension.
<code>outputs/</code>	Generated tables, figures and log reports.
<code>outputs_large/</code>	Generated tables, figures, quality-control reports and summary outputs for the large-network extension.
<code>network_exports/</code>	Reusable node tables and edge lists for both the original 47-stock network and the 474-stock extension at the main threshold.

Table B.2: Software environment recorded for the real-data run.

Component	Version
Python	3.12.4
pandas	3.0.5
NumPy	2.5.2
NetworkX	3.6.1
Matplotlib	3.11.1
seaborn	0.13.2
yfinance	1.6.0
SciPy	1.18.1

B.3 Reproducibility Procedure

The empirical results can be regenerated from the project root using the dependencies and execution commands provided with the repository. After obtaining the source code, the required Python packages are installed with:

```
pip install -r requirements.txt
```

The original 47-stock real-data analysis used for the main empirical study is executed with:

```
python main.py --refresh
```

The supplementary 474-stock large-network extension is executed separately with:

```
python large_network_extension.py --refresh
```

Once started, `main.py` automatically executes the complete empirical pipeline: data collection, calendar alignment and preprocessing, computation of logarithmic returns, estimation of the Pearson correlation matrix, construction of the threshold-filtered graphs, Breadth-First Search, Depth-First Search, connected-component analysis, articulation-point and bridge analysis, construction of the Minimum Spanning Tree, calculation of network metrics, generation of the exploratory portfolio comparison, and export of the resulting tables, figures and log reports.

The main parameters governing the analysis—including the sample period, stock universe, correlation thresholds, main threshold and BFS source stock—are declared in `config.py`. The command-line interface in `main.py` also allows the sample period, threshold grid, main threshold and BFS source to be overridden for an individual run without modifying the configuration file.

The large-network script applies the same core correlation and threshold-network framework to the broader stock universe, while adding sector-alignment analysis, large-network quality-control checks and dedicated exports. Its outputs are written separately from those of the original pipeline so that the two empirical experiments cannot be mixed accidentally.

The project additionally provides a synthetic self-test mode for checking whether the pipeline executes correctly without downloading market data. Synthetic outputs are stored separately and are not valid thesis results. For every run, the pipeline generates a data-source report identifying whether the analysis used real Yahoo Finance data or simulated data. Only outputs identified as real Yahoo Finance data are used in this thesis.

B.4 Output Files

The pipeline exports its results as CSV tables, PNG figures at 300 DPI and text-based log reports. The main output categories are:

- correlation matrix outputs, including the full Pearson correlation matrix and the sector-average correlation matrix;
- threshold network statistics, including edges, density, degree, clustering, components and giant-component size across the threshold grid;
- BFS results, including distances, layers and reachability from the configured source stock;
- DFS and articulation-point outputs, including connected-component recovery, articulation points and bridges;
- MST outputs, including tree edges, node-level tree metrics and sector-branch composition;
- portfolio comparison outputs, including risk and return statistics and holdings for the illustrative portfolios;
- generated figures, including correlation heatmaps, threshold-network diagrams, BFS visualisations, the MST diagram and portfolio comparison charts.
- large-network threshold statistics, including nodes, edges, density, average degree, clustering, connected components, giant-component size, isolated nodes, same-sector edge share and sector assortativity;
- node-level and sector-level alignment outputs for the 474-stock extension, including same-sector neighbour shares, dominant neighbour sectors and the descriptive cross-sector-oriented classification;
- reusable network exports stored in `network_exports/`, containing node tables and edge lists for the original and large networks at $\tau = 0.5$;
- large-network provenance and quality-control reports documenting ticker retention, metadata sources and validation checks.

B.5 Selected Core Implementations

The complete source code is available in the accompanying GitHub repository, with an archived copy of the thesis version provided as `thesis_code.zip`. This section reproduces only the core traversal routines most directly related to the computer-science contribution

of the thesis. The listings are imported directly from the original project files rather than retyped or modified here.

B.5.1 Custom Breadth-First Search

Listing B.1 reproduces the custom Breadth-First Search implementation used in the empirical pipeline. The routine maintains an explicit FIFO queue, records the visitation order, shortest-path distance and predecessor of every reachable node, and visits neighbours in sorted order for deterministic execution.

Listing B.1: Original custom Breadth-First Search implementation from `src/traversal_analysis.py`.

```

1 def custom_bfs(graph: nx.Graph, source: Hashable
2     ) -> Tuple[List[Hashable], Dict[Hashable, int], Dict[Hashable, Hashable
3     | None]]:
4
5     """Breadth-First Search implemented from first principles.
6
7     The algorithm maintains a FIFO queue of discovered-but-unexplored nodes.
8     Because every edge has unit cost in the *unweighted* threshold graph, the
9     order in which BFS dequeues nodes is exactly the order of increasing
10    distance from the source, which is why BFS also solves the shortest-path
11    problem here.
12
13    Neighbours are visited in alphabetical order so that the traversal is
14    deterministic and reproducible across runs and machines.
15
16    Complexity:  $O(|V| + |E|)$  time,  $O(|V|)$  space.
17
18    Returns
19    -----
20    (order, distance, parent)
21        'order' nodes in the order they were dequeued;
22        'distance' number of edges from the source (missing = unreachable);
23        'parent' predecessor of each node in the BFS tree.
24    """
25    if source not in graph:
26        raise KeyError(f"Source node {source!r} is not in the graph.")
27
28    visited = {source}
29    distance: Dict[Hashable, int] = {source: 0}
30    parent: Dict[Hashable, Hashable | None] = {source: None}
31    order: List[Hashable] = []
32    queue: deque = deque([source])
33
34    while queue:
35        node = queue.popleft()          # FIFO -> breadth first

```

```

34     order.append(node)
35     for neighbour in sorted(graph.neighbors(node)):
36         if neighbour not in visited:
37             visited.add(neighbour)           # mark on discovery, not on visit,
38             distance[neighbour] = distance[node] + 1 # so no node is queued
39         parent[neighbour] = node
40         queue.append(neighbour)
41     return order, distance, parent

```

B.5.2 Custom Depth-First Search

Listing B.2 reproduces the iterative custom Depth-First Search routine. As discussed in Chapters 3 and 4, the exact DFS parent tree is not used as an empirical result; the routine is used for traversal coverage, while articulation points and bridges are computed separately with NetworkX's Tarjan-based routines.

Listing B.2: Original custom Depth-First Search implementation from `src/traversal_analysis.py`.

```

1  def custom_dfs(graph: nx.Graph, source: Hashable
2      ) -> Tuple[List[Hashable], Dict[Hashable, Hashable | None]]:
3      """Depth-First Search implemented iteratively (pre-order).
4
5      An explicit LIFO stack is used instead of recursion: the recursion depth of
6      a natural recursive DFS is bounded by  $|V|$ , which is safe for 47 stocks but
7      not for a larger universe, and an explicit stack makes the "go deep, then
8      backtrack" behaviour visible in the code.
9
10     Neighbours are pushed in reverse alphabetical order so that they are popped
11     alphabetically, matching the deterministic convention used in 'custom_bfs'.
12
13     Complexity:  $O(|V| + |E|)$  time,  $O(|V|)$  space.
14     """
15     if source not in graph:
16         raise KeyError(f"Source node {source!r} is not in the graph.")
17
18     visited = set()
19     parent: Dict[Hashable, Hashable | None] = {source: None}
20     order: List[Hashable] = []
21     stack: List[Hashable] = [source]
22
23     while stack:
24         node = stack.pop()           # LIFO -> depth first
25         if node in visited:
26             continue                # already reached by a deeper branch

```

```

27         visited.add(node)
28         order.append(node)
29         for neighbour in sorted(graph.neighbors(node), reverse=True):
30             if neighbour not in visited:
31                 parent.setdefault(neighbour, node)
32                 stack.append(neighbour)
33     return order, parent

```

B.5.3 DFS Forest and Connected-Component Recovery

The full DFS forest restarts the custom DFS from each not-yet-visited root so that all connected components of a possibly disconnected threshold graph are covered.

Listing B.3: Original DFS-forest implementation used for full-graph traversal and connected-component recovery.

```

1 def custom_dfs_forest(graph: nx.Graph) -> Tuple[List[Hashable], Dict[Hashable,
2     Hashable | None],
3     List[List[Hashable]]):
4     """Run DFS from every not-yet-visited node to cover the whole graph.
5
6     A single DFS only reaches the component of its source. Restarting on every
7     unvisited node produces a DFS forest whose trees are exactly the
8     connected components of the graph - which is the classical way of computing
9     connected components with a traversal, and the way it is presented in
10    Chapter 3.
11    """
12    visited = set()
13    order: List[Hashable] = []
14    parent: Dict[Hashable, Hashable | None] = {}
15    components: List[List[Hashable]] = []
16
17    for root in sorted(graph.nodes()):
18        if root in visited:
19            continue
20        local_order, local_parent = custom_dfs(graph, root)
21        visited.update(local_order)
22        order.extend(local_order)
23        parent.update(local_parent)
24        components.append(local_order)
25    return order, parent, components

```

B.5.4 BFS Validation Against NetworkX

For BFS, shortest-path distances are canonical, so the custom implementation can be checked directly against NetworkX. Listing B.4 reproduces the exact validation routine

used in the project.

Listing B.4: Original BFS validation routine from `src/traversal_analysis.py`.

```
1 def validate_bfs(graph: nx.Graph, source: Hashable,
2                 order: Sequence[Hashable], distance: Dict[Hashable, int]) -> Dict[str,
3                 bool]:
4     """Check the custom BFS against NetworkX's reference implementation.
5
6     Two properties are checked:
7
8     * the set of reached nodes equals the source's connected component, and
9     * every distance equals 'nx.single_source_shortest_path_length'.
10
11    BFS distances are canonical (they do not depend on tie-breaking), so an
12    exact match is expected and any mismatch is a genuine bug.
13    """
14    reference_distance = dict(nx.single_source_shortest_path_length(graph, source))
15    reachable_reference = set(reference_distance)
16
17    checks = {
18        "same_reached_set": set(order) == reachable_reference,
19        "same_distances": distance == reference_distance,
20        "order_is_non_decreasing": all(distance[a] <= distance[b]
21                                       for a, b in zip(order, order[1:])),
22    }
23    for name, ok in checks.items():
24        logging.log(logging.INFO if ok else logging.ERROR,
25                    "    BFS validation - %-24s : %s", name, "PASS" if ok else "FAIL")
26    return checks
```

The repository also contains the project's DFS diagnostic routine. It is not reproduced here because, as stated in Chapter 3, the thesis does not treat those checks as an independent proof of every possible DFS-ordering property. The substantive DFS-related empirical conclusions rely on connected-component recovery and on the separate Tarjan-based articulation-point and bridge routines.

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